GARBAGE MANAGEMENT SYSTEM

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The above candidate has carried out work for the Bachelors of Information Technology Degree report under my supervision.

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Abstract

Most developing countries face significant challenges in garbage management, resulting in environmental pollution, spread of disease, climate change impacts, and resource wastage. In Sri Lanka the responsibility for garbage management lies with municipal councils, however many municipal councils lack on resources, and continue to rely on manual and outdated methods. These methods include paper based records, improper scheduling, and manual complaint handling which leads to poor coordination, lack of transparency, and public dissatisfaction.

This project aims to develop a centralized system to digitize garbage collection process. The proposed system supports communication among resident, administrator, truck drivers. Key features including real time truck tracking, QR code base bin identification & status updates, effective scheduling, digital complaint tracking, notifications service and public awareness features. The system is planned to develop using react.js for front end, node.js for backend, MySQL as the database, and Google maps for truck location tracking. The development process follows a structured methodology, including requirement analysis, system design, implementation, and testing phases ensuring final product meets both functional and non functional requirements.

The expected results of the proposed system include improved operational efficiency, enhanced transparency, better communication among stakeholders, and increased public satisfaction. Overall CleanLand garbage management system offers a practical and scalable solution for modernize garbage management in Sri Lanka, contributing to a cleaner environment and promoting sustainable garbage management practices.

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List of abbreviations

MT- Matric Ton

GPS- Global Positioning System

QR- Quick Response

MC- Municipal Council

CMC- Colombo Municipal Council

1.Introduction

1.1. Background

Garbage management is an essential public service that directly impacts environmental sustainability, public health and overall quality of life in the urban and suburban environments. As there is increasing population growth and urbanization, there is significant generation of garbage by residents and commercial sectors.

Most of the developing countries face the problem of managing waste properly. In 2012, cities world over, generated 1.3 billion tons of solid waste per year, amounting to a footprint of 1.2 kilograms per person, per day needless to state that with rapid population growth and urbanization, the municipal waste generation is expected to increase to 2.2 billion tons (MT) by 2025. Inadvertently, with the current trends continuing, it is likely to rise from 3.5 MTs to 6 MTs per day with, each person generating around 0.64 kg waste per day in Sri Lanka with an estimated 4.8 billion MT of waste collected per annum in the country. [1]

Municipal councils are responsible for ensuring timely and efficient waste collection, however in Sri Lanka, garbage collection process are still handled manually. These traditional methods often lead to inefficiencies, lack of transparency, poor coordination, and dissatisfaction.

Presently, most garbage collection activities rely on improper schedules, paper based documents, telephone communications, and physical complaints lodged in the municipal offices. Residents don't even aware of garbage collection times, bin types & updates, and route updates leading to missed pickups and overflowing bins. At the same time municipal authorities face difficulties in tracking garbage trucks, managing complaints, and responding promptly to service issues. The absence of a centralized digital platform causes delay, human errors, data loss, and limited accountability in garbage management operations

Advancement in web technologies, GPS tracking capabilities, QR code systems, and complaint management provide a chance to improve garbage collection management services. By developing a web based garbage collection solution, it is possible to bring residents, garbage collectors/ drovers, and admin on a single platform and it improves coordination and communication among all stakeholders. Features such as real time truck tracking, online complaint management, garbage collection scheduling, bin identification and bin management using QR code can significantly enhance service efficiency and transparency.

The proposed CleanLand Garbage Management System is designed to address these issues by replacing manual process with an automated and centralized web based system. The

main goal of the system aims to streamline garbage collection operations, provide real time information to residents, support municipal councils in monitoring and decision making and promote eco friendly garbage disposal practices. By improving awareness, accountability, and responsiveness, the system contributes to building cleaner cities, reducing environmental pollution, and enhancing public trust in municipal services

1.2. Problem Statement

Sri Lanka as a country with developing economy finds it challenging to impose a financially, environmentally and socially acceptable waste management solution. Currently Sri Lanka generates 7000 Metric Ton of solid waste per day but only about half of that waste is properly collected leading to serious environmental and public health concerns.[2]

Most municipal council areas in Sri Lanka rely on manual and outdated methods for collecting methods. These process mainly involve improper schedule, paper based records, phone communications, and in person complaint handling, which is highly inefficient and difficult to track. As a result, residents, drivers, and municipal council administrators face frequent challenges in coordinating and managing garbage collection services effectively.

The main problem is the lack of a centralized, digital system that connects all stakeholders involved in garbage management. Residents often do not have timely information about garbage collection schedules, garbage truck locations, or proper waste disposal methods. Complaints related to missed pickups, overflowing bins, and irregular services are usually reported manually, leading to delays, poor follow up, and dissatisfaction. At the same time, municipal councils have limited visibility into real time garbage truck movements, bin status, route progress, and complaint resolution, making it difficult to ensure quality and accountability

This problem exists across municipal council areas, particularly in urban and semi urban regions, where waste generation is high and efficient collection is critical. The issue occurs on a daily basis, as garbage collection is a continuous public service that requires regular monitoring and coordination. The absence of real time data and automated tools results in operational inefficiencies, miscommunication, data loss, and reduced transparency.

The root cause of these issue is the reliance on manual processes and disconnected systems, which do not support real time tracking, proper scheduling, digital complaint handling, or performance monitoring. Without a modern, web based solution, municipal council struggle to meet the increasing demands of garbage management, and residents are unable to actively participate in maintaining a clean and healthy environment

Therefore, there is a clear need for a centralized, web based garbage management system that bridges the gap between the current inefficient state and the desired goal of efficient,

transparent, and sustainable garbage collection process. The system should enhance communication, allow real time monitoring, be efficient in operations, and contribute to increased public satisfaction while encouraging eco friendly waste management practices.

1.3. Motivation & Significance of Project

The motivation for developing the CleanLand Garbage Management System arises from the growing challenges faced by municipal councils and residents in managing garbage collection efficiently. With increasing urbanization and population growth, the volume of waste generated daily has significantly increased, placing pressure on existing garbage collection systems. However, most of the municipal authorities are using manual and inefficient methods of garbage collection, which are no longer able to meet the rising demands in the field of garbage management.

From a practical perspective, inefficient garbage management leads to serious consequences such as environmental pollution, public health risks, unpleasant living conditions and reduced public trust in municipal services. Missed garbage pickups, overflowing bins, and delayed complaint resolution not only inconvenience residents but also contribute to the spread of diseases and environmental degradation. The lack of real time information and proper monitoring tools prevents municipal authorities from responding actively to issues and optimizing their resources.

The relevance of this project lies in the fact that it has the ability to convert the traditional process of garbage collection into a digital, data driven system for garbage management. By introducing features such as real time garbage truck tracking, complaint management, QR code based bin identification and update, scheduled collection updates, and email notifications, the system enhances communication between residents, garbage collectors, and municipal administrators. This integrated approach improves accountability, reduces operational inefficiencies, and ensures timely service delivery

From a research and academic perspective, this project demonstrates the application of modern web technologies such as React.js, Node.js, MySQL, GPS based tracking, and QR code integration to solve real world problems. It contributes to the growing body of research on smart city solutions and sustainable garbage management by proposing a scalable and adaptable digital platform. It contributes to the growing body of research on smart city solutions and sustainable waste management by proposing a scalable and adaptable digital platform.

Overall, the CleanLand garbage management system is significant because its not only addresses an existing operational problem but also supporting long term environmental sustainability, public health improvement, and smart governance. By digitizing garbage management processes, the system helps bridge the gap between residents and municipal

councils while promoting responsible garbage disposal practices and contributing to the development of cleaner and healthier communities.

1.4. Aims and Objectives

1.4.1 Aim

The aim of this project is to design and develop a centralized, web based garbage management system that will make service of municipal garbage collection more efficient, transparent, and reliable. The system seek to bridge the gap between residents, garbage drivers, and municipal council administrators by providing real time truck information, effective communication, and improved service monitoring for cleaner environments and sustainable waste management practices.

1.4.2 Objectives

- Enhance communication and coordination among residents, garbage collectors, and municipal council administrator on a single digital platform.
- Increase transparency in garbage collection operations by enabling real time tracking of garbage trucks and insight into collection schedules.
- Improve service efficiency by reducing missed pickups, delayed collections, and unmanaged garbage bins through effective monitoring and reporting
- Provide residents with convenient access to garbage collection information, complaint submission, feedback mechanism, and email notifications.
- Support municipal councils with the monitoring operations, managing complaints, tracking, and making decisions. through the use of accurate and timely data.
- Promote responsible and eco friendly garbage disposal practice by increasing public awareness and providing proper guidance on garbage management.
- Establish a scalable and adaptable system that can support future enhancement and contribute to smart city and sustainable development initiatives.

2.Literature Review

This chapter presents a comprehensive analysis of existing systems pertaining to garbage management in Sri Lanka and around the world. An extensive analysis has been done on these

systems, identifying their strengths and weaknesses, their technological implementation, the system function and how they contribute to efficiency handling garbage management.

A survey also done among the system stakeholders to gain further understanding of the existing system, their limitations and pain points, and to understand the scope for further improvement.

The literature review has helped us identify the gaps and strengths in the existing systems, thereby giving us foundational guidelines for developing our Garbage Management System.

2.1. Critical evaluation of 5 existing solutions

2.1.1. MYWASTE Garbage Collection System

The MYWASTE [6] platform is a cloud based digital waste management solution design to support comprehensive waste traceability, operational control and reporting throughout the waste lifecycle. The system provides tools for waste generation, and mobile collection registration with location and photographic evidence. Although primarily targeted at business and waste transport operates, many of its features align with modern digital waste management objectives.

Despite its robust operational capabilities, MYWASTE focuses on enterprise users rather than individual residents. It offers high visibility into waste logistics and supports regulatory compliance through automated documentation, but lacks fully integrated public engagement features such as user complaint tracking, personalized notifications, and user feedback dashboards. Consequently, while user satisfaction is high among business operators and logistics managers, its fulfillment of broader community centric requi/rements is moderate. The absence of tools for direct resident interaction highlights opportunities for more inclusive systems like the proposed CleanLand platform, which aims to combine operational efficiency with accessible digital engagement for citizens.

Components of the system

1.Garbage control and tracking

- This component tracks waste from generation to final destination, provides complete tracking

2.Real time collection & route tracking

- Create optimized collection routes and real time GPS monitor of trucks

3.Automated document generation

- System automatically generates required garbage transport documents

4.Collection registration

- Drivers and field workers can register collection using GPS location, digital signature and photos

Advantages & Disadvantages

Component	Advantage	Disadvantage
Garbage tracking	Full life cycle tracking	Requires proper data entry
Reak time routing	Efficiency and visibility	Need GPS & connectivity
Automated documentation	Compliance support	Training required
Mobile collection app	Digital proof	Tools dependency

Table 1: Advantage & disadvantages of MYWaste garbage management system

Conclusion

The MYWASTE platform offers a enterprise level solution that enhances waste traceability regulatory compliance through advanced digital tools.it achieves high user satisfaction among business operators, it align with broader municipal and resident requirements remain moderate. The absence of public engagement features highlights the need for future systems, such as CleanLand, to balance operational efficiency with accessible public services.

2.1.2. Cleantech Garbage Management System

The Cleantech Solid Waste Management service page [7] describes a suite of environmental and waste management solutions offered by Cleantech (Pvt) Ltd, including garbage collection, recyclable material collection, mobile sanitation, urban beautification, awareness campaigns, and waste management planning for residential and commercial clients. The service emphasize on ground operational support intended to improve cleanliness, hygiene, and community participation in waste practices.

While Cleantech's services cover a broad range of waste management activities and contribute to improved environmental outcomes, the system lacks a digital platform for user interaction. There are no real time collection schedules, online complaint tracking, automate notifications, or interactive dashboards available for residents through the website. As a result, access to services remains largely manual and passive. Although the operational scope is wide, user satisfaction and requirement fulfillment for modern, digital

systems such as the proposed CleanLand platform that product interactive user engagement.

Components of the system

1.Garbage collection management

- Collect, transport, and manage household and commercial garbage

2.Recyclable collection service

- Collection and processing of recyclables such as plastic, paper, metal, and e waste.

3.Mobile toilets

- Portable sanitation solutions including mobile toilets and hygiene services.

4.City beautification

- Enhancement of public spaces, landscaping and beautification initiatives

5.Awareness campaigning

- Community outreach to educate citizens on responsible waste handling proper disposal

6.Waste management planning

- Consultation service for designing waste management frameworks for building and apartments.

Advantages & Disadvantages

Component	Advantage	Disadvantage
Garbage collection management	Reduce accumulation and cleanliness	Lack of schedules
Recyclable collection service	Easy to request for recycle	User cannot track recycling request digitally
Awareness campaign	Supports source segregation and community engagement	Does not provide personalized or interactive learning tools
Waste management planning	Valuable for large residential and commercial clients	Clients may not receive real-time monitoring dashboards

Table 2: Advantage & disadvantages of Clean Tech garbage management system

Conclusion

Cleantech garbage management provides effective on ground services that enhance urban cleanliness and environmental hygiene. However, the absence of digital interaction, real time tracking, and user engagement limits user satisfaction. This highlights the need for integrated digital systems to improve transparency, accessibility, and citizen involvement in modern waste management.

2.1.3 Colombo Municipal Council Garbage Collection System

The Colombo municipal council's web based platform provides a waste management sub page [3] that provides area wise garbage collection schedules for Colombo city. The system displays collection routes, road level frequencies, and limited operational details such as contractor information and supervisor contact numbers. All information is provided in static text table formats.

Although the system improves public access to official waste collection information, it lacks real time tracking, notifications, real time updates, and interactive complaint handling. Users must manually search schedules and contact authorities outside the system for inquiries or complaints. As a result, user satisfaction and requirement fulfillment remain low compared with modern digital garbage management solutions such as proposed system**.**

Components of the system

1. Collection schedule by area
    - List specific areas(D1, D2A, D2B)
    - Shows roads with garbage collection frequencies (once/day, twice a day)
2. Contractors & supervisors
    - Identifies collection contractors for some areas with supervisor contact numbers.
3. Time and labor details

- Some areas include details like start end time of collection, contact numbers, and days of collection for selected roads

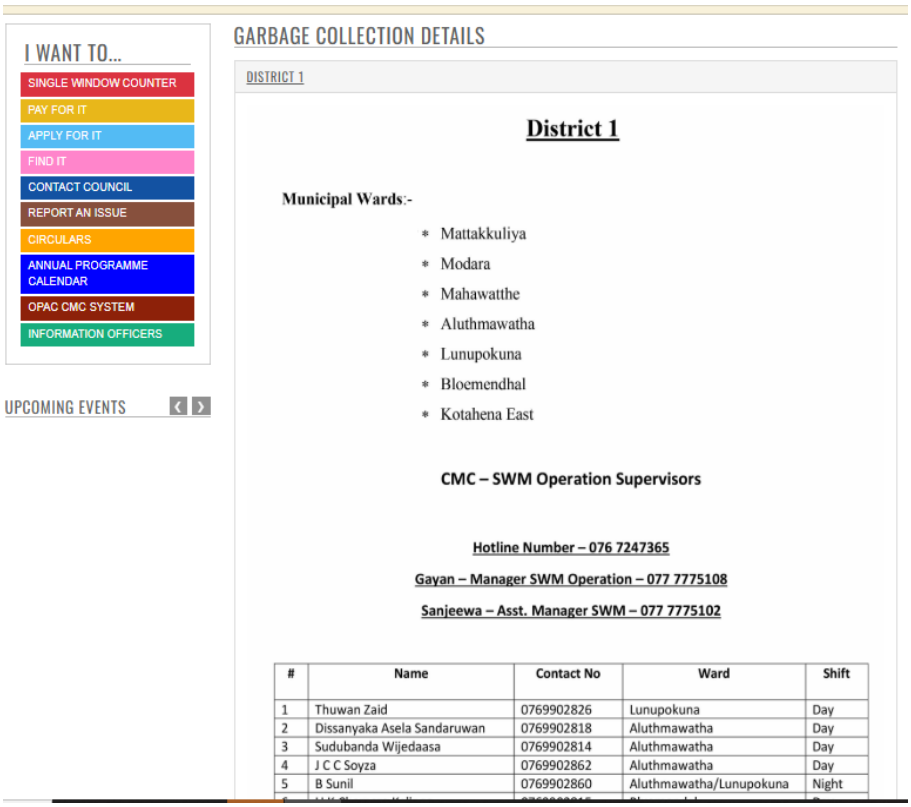
4. Public information
   - There are static text listings of schedules and routes
5. Complaints
   - Area wise mobile contact numbers are available for submitting complaints and inquiries

Advantages & Disadvantages

Component	Advantage	Disadvantage
Collection schedules	Provided area & road wise schedules in details	Static table format it's difficult to find specific areas quickly, not interactive
Communication	Gives contact details for supervisors which can be useful	No mobile link or call automation routing, user must manually copy numbers
Schedules	Some areas provide "from- to" timings	Confusing format and not consistently available for all districts
Availability	Accessible via official website, trusted	No notification, no complaint facility on the same page

Table 3: Advantage & disadvantages of CMC garbage management system

Existing system interfaces



I WANT TO...

SINGLE WINDOW COUNTER
PAY FOR IT
APPLY FOR IT
FIND IT
CONTACT COUNCIL
REPORT AN ISSUE
CIRCULARS
ANNUAL PROGRAMME CALENDAR
OPAC CMC SYSTEM
INFORMATION OFFICERS

UPCOMING EVENTS

GARBAGE COLLECTION DETAILS

DISTRICT 1

District 1

Municipal Wards:-

* Mattakkuliya
* Modara
* Mahawatthe
* Aluthmawatha
* Lunupokuna
* Bloemendhal
* Kotahena East

CMC – SWM Operation Supervisors

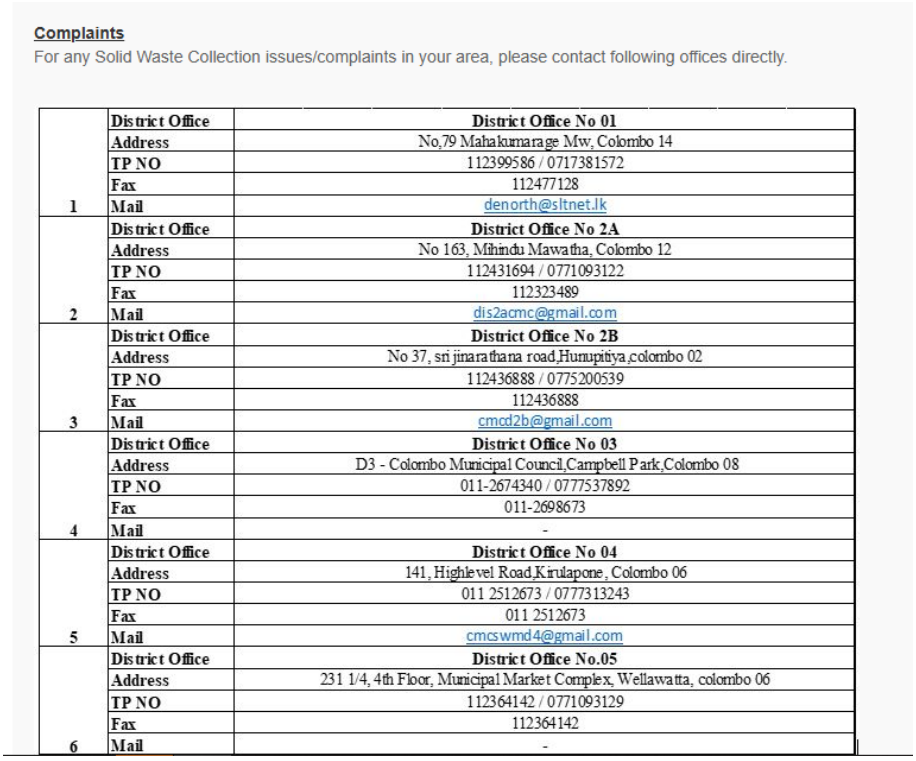
Hotline Number – 076 7247365

Gayan – Manager SWM Operation – 077 7775108

Sanjeewa – Asst. Manager SWM – 077 7775102

#	Name	Contact No	Ward	Shift
1	Thuwan Zaid	0769902826	Lunupokuna	Day
2	Dissanyaka Asela Sandaruwan	0769902818	Aluthmawatha	Day
3	Sudubanda Wijedaasa	0769902814	Aluthmawatha	Day
4	J C C Soyza	0769902862	Aluthmawatha	Day
5	B Sunil	0769902860	Aluthmawatha/Lunupokuna	Night

Figure 1:CMC Existing system interfaces



Complaints

For any Solid Waste Collection issues/complaints in your area, please contact following offices directly.

1	**District Office**	**District Office No 01**
	Address	No,79 Mahakumarage Mw, Colombo 14
	TP NO	112399586 / 0717381572
	Fax	112477128
	Mail	denorth@sltnet.lk
2	**District Office**	**District Office No 2A**
	Address	No 163, Mihindu Mawatha, Colombo 12
	TP NO	112431694 / 0771093122
	Fax	112323489
	Mail	dis2acmc@gmail.com
3	**District Office**	**District Office No 2B**
	Address	No 37, sri jinarathana road,Hunupitiya,colombo 02
	TP NO	112436888 / 0775200539
	Fax	112436888
	Mail	cmcd2b@gmail.com
4	**District Office**	**District Office No 03**
	Address	D3 - Colombo Municipal Council,Campbell Park,Colombo 08
	TP NO	011-2674340 / 0777537892
	Fax	011-2698673
	Mail	-
5	**District Office**	**District Office No 04**
	Address	141, Highlevel Road,Kirulapone, Colombo 06
	TP NO	011 2512673 / 0777313243
	Fax	011 2512673
	Mail	cmcswmd4@gmail.com
6	**District Office**	**District Office No.05**
	Address	231 1/4, 4th Floor, Municipal Market Complex, Wellawatta, colombo 06
	TP NO	112364142 / 0771093129
	Fax	112364142
	Mail	-

Figure 2: CMC Existing system interfaces -Contact information

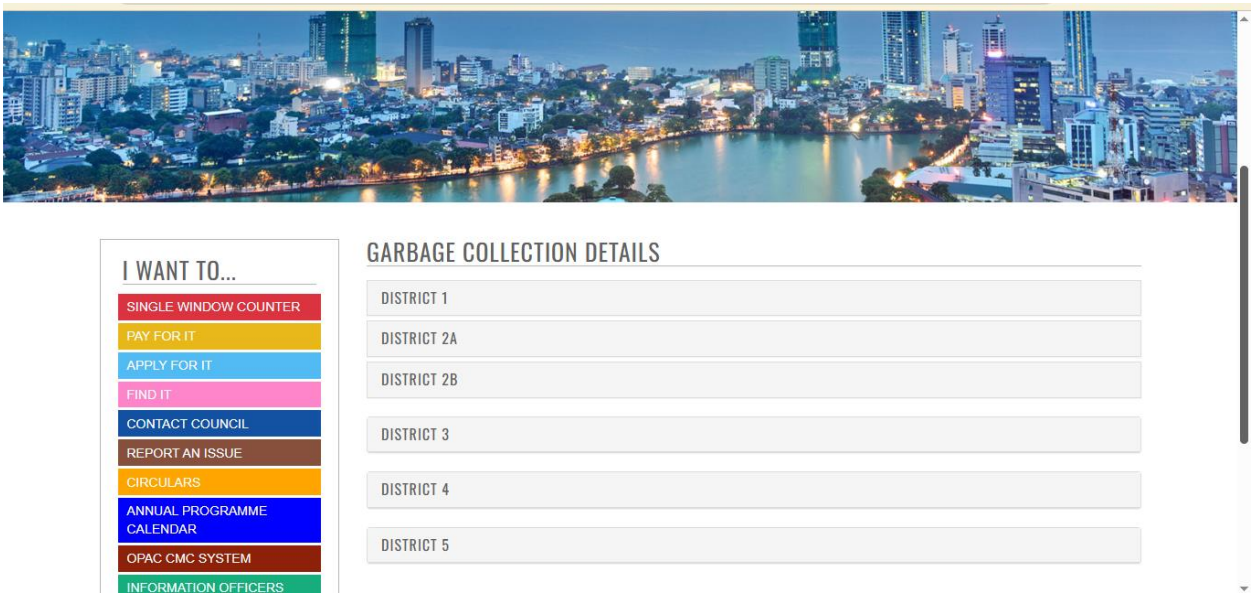


Figure 3: CMC Existing system interfaces

Conclusion

The existing CMC garbage collection systems mainly function as informational platforms. They lack interactive features such as real time tracing, digital complaint handling, and user notification. Communication between residents and waste management authorities is limited and inefficient. Therefore, these systems do not fully satisfy modern user and service requirements, emphasizing the need for an improved digital solution.

2.1.4 Kurunegala Municipal Council Garbage Collection System

The Kurunagalle Munciple Council's garbage management page is a web based informational platform [4] that provides residents with garbage collection schedules, administrative division details, and council services such as yellow bag collection and dustbin hiring. Users can view collection schedules by secret division divisions and submit complaints through a basic online form. Static notices and service listings are also included

While the system make official waste management information publicly accessible, it remains primarily descriptive and lacks interactive or real features. The complaint form has experienced submission issues, and no status tracking or notification system are available. Due to these limitations, user satisfaction is low and requirement fulfillment is minimal compared to modern waste management platforms. The absence of digital engagement,

real time updates, and feedback mechanism highlights the need for more integrated solutions such as the proposed system.

Components of the system

1. Collection schedule management
   - Displays garbage collection schedule by day of the week & division
2. Notice
   - Static space for notice

4. Complaint management
   - Form is available for submitting complaints and inquiries but there are some issues is submitting
5. Municipal service management
   - Include services such as yellow bag collection, dustbin hiring.
6. Division module
   - Shows GN divisions, PHI zones, and council jurisdictions.

Advantages & Disadvantages

Component	Advantage	Disadvantage
Collection schedules	Provided division (GN) based fixed schedule	Lack of real time update & "No record found error"
Division modules	Clearly defined service areas, & structured division listing	No interactive map navigation, & static representation
Service management	Support special waste handlings	No online bookings or follow ups
Complaint management	Enables public participation, & user friendly complaint form	No status update & server side error observed

Table 4: Advantage & disadvantages of Kurunegala MC garbage management system

Existing system interfaces



Figure 4: Kurunegala MC existing system interface

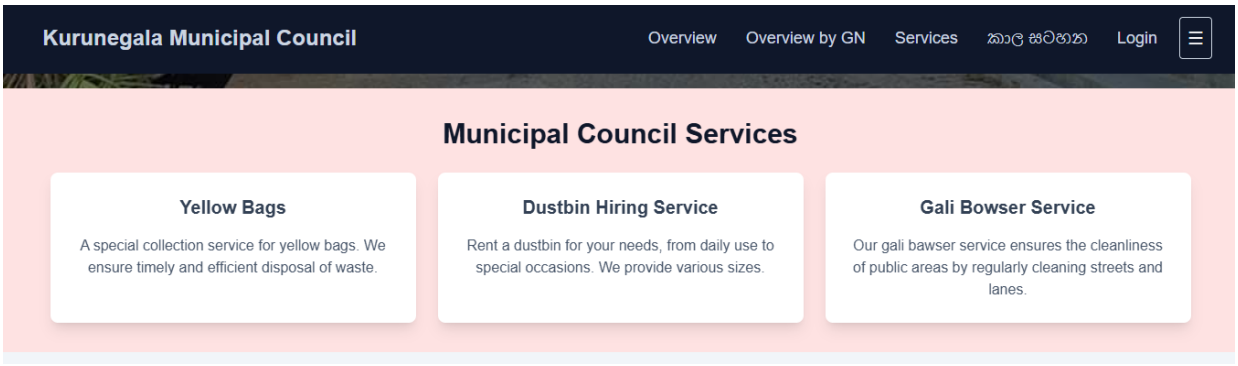


Figure 5: Kurunegala MC Existing system interfaces

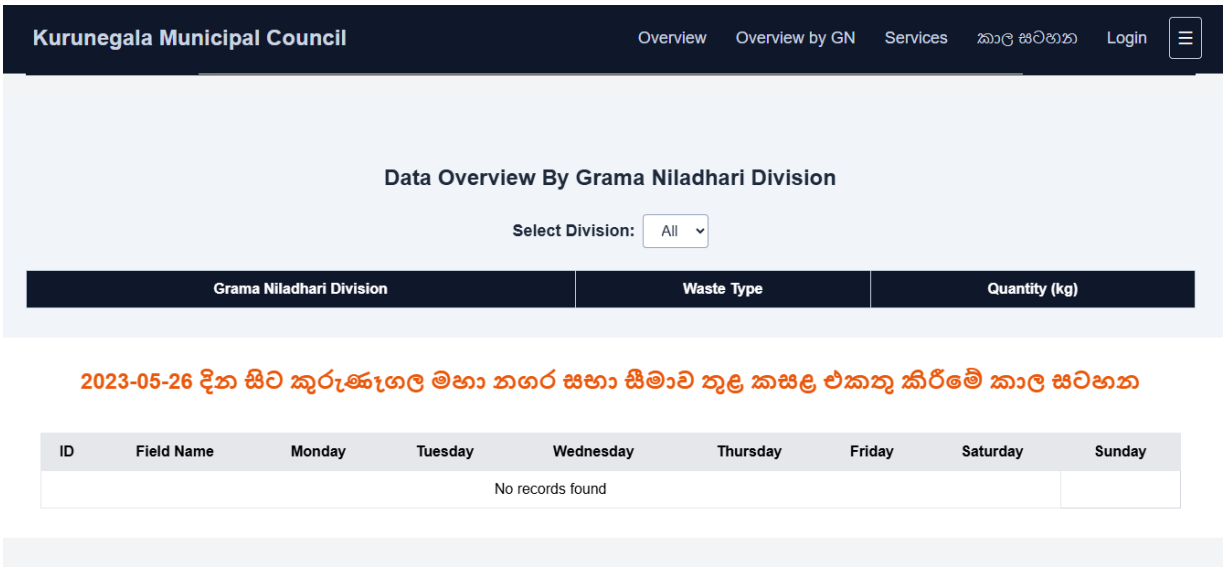


Figure 6:Existing system interfaces-Schedules

2.1.5 Galigamuwa Municipal Council Garbage Collection System

The Galigamuwa Municipal council garbage management page [5] provides essential information regarding planned garbage collection route, community waste handling practices, and public education materials related to solid waste. The page lists collection routes and areas and offers Educational content on garbage waste definitions, 3R concepts, and proper segregation is also included, along with general details on resources deployment, composting activities, and recycling processes. contract directions for submitting complaints or feedbacks.

Although the system improves transparency and raises awareness of waste management responsibilities within the community, it remains largely static and informational. Collection schedules are not clearly presented with specific times, resource details lack real time status, and public education content does not support user intention. Complaint procedures exist but are not integrated with digital tracking or status updates. As a result, user satisfaction and requirement fulfillment are moderate at best. These limitations emphasize the need for a more interactive, real time, and responsive solution such as the proposed system

Components of the system

1. Garbage collection routes and schedules
   - List collection rotes and areas
   - Dedicated WhatsApp group for route share

2. Human and physical resource deployment

- Supports internal resource planning and transparency in workforce deployment by providing number of vehicles, drivers and laborers allocated per route

3. Waste processing and recycling information
   - Show garbage is collected, processed and managed sustainably by giving compost details
4. Public education on garbage and 3R concept
   - Static text explanation on what solid waste is, 3R concepts, environmental and health issues of improper waste disposal, and waste segregation
5. Complaint management
   - Not directly detailed on the waste management page but council citizen chart indicates acknowledge of complaints within 7 days, final decision within 21 days, and contact details for feedback submitted or via phone
   - Contact and quick reply through WhatsApp.

Advantages & Disadvantages

Component	Advantage	Disadvantage
Collection schedules	Resident can know where collection occurs Provide area based schedules	Schedules are not clearly displayed as time
Resources	Shows resource allocation and accountability	Lack of real time availability, hard understand
Waste process	Show environmental commitment	No schedule or progress displayed
Public Education	Educate the community	No user interaction, not user friendly
Complaint & response	Timed acknowledged process	No tracking or notifications on status

Table 5: Advantage & disadvantages of Galigamuva garbage management system

Conclusion

This shows that garbage management system mainly function as informational platforms with limited interactivity. Although they improve transparency and public awareness, they fail to support real time operations, and user engagements. user satisfaction and requirement fulfillment remain moderate to medium These gaps justify the need for the improved Garbage Management System to provide a more integrated, scalable, and user-centric solution.

2.1.6. Galle Municipal Council Garbage Management System

The Galle Municipal Council provides page [8] for garbage management services, page outline the primary responsibilities and procedures for garbage management services by the municipal council. Page highlights key functions such as garbage removal, town cleaning activities, complaint reporting, and implementation of recycling programs. Procedures include logging complaints via telephone, sorting collected waste for recycling, and removal of town garbage according to a time table.

While the page provides an official overview of the council's garbage management roles and responsibilities, it remains largely descriptive and lacks digital interactivity or user facing features. There are no web based schedules for garbage collection, real time updates on waste operations, online complaint submission forms, or dashboards for residents. As such, user satisfaction is limited by the absence of accessible digital services, and requirement satisfaction is minimal in terms of modern garbage management expectations. Although the content increases transparency about organizational tasks, the lack of a structured interactive platform underscores the need for an integrated system like the proposed system, which supports real time tracking, notification and enhanced citizen engagement.

Components of the system

1. Waste management services overview
2. Resident complaint report service
3. Recycle & waste sorting process
4. Town cleaning & garbage removal service

Advantages & Disadvantages

Component	Advantage	Disadvantage
Waste management services overview	Clearly explains the scope of the services provided by municipal council	Information is descriptive only, no interaction with residents
Complaint report service	Allow residents to report complaints through call	No online complaint submission, no tracking for complaints
Recycle & waste sorting process	Demonstrates environmental responsibility and recycling practices	No information on recycling schedules
Town cleaning & garbage removal	Ensures routine cleanliness and daily waste removal activities	Collection schedules are not published online in detail for residents

Table 6: Advantage & disadvantages of Galle MC garbage management system

Comparison

	Colombo Municipal Council	Kurunegala Municipal Council	Galigamuwa Municipal Council	MYWASTE	Cleantech	Galle Municipal Council	CleanLand
Collection schedule	Yes	Yes (incomplete)	Yes (incomplete)	-	No	No	Yes
Real time truck tracking	No	No	No	Yes	No	No	Yes
Complaint management	No	Yes	Yes	Yes	No	Yes	Yes
Notification	No	No	No	No	No	No	Yes
Special request	Yes	Yes	No	Yes	Yes	No	Yes
QR based bin identification	No	No	No	NO	No	No	Yes
Public feedback and rating	No	No	No	No	N0	No	Yes

Table 7: Comparison with existing systems

2.1.7 Background analysis and survey among different stakeholders

Stakeholder survey was conducted to evaluate existing garbage collection system and identify user expectations for the proposed system. The survey was conducted through Google forms during December 2024 and received responses from 36 participants representing urban (27.8%), suburban (52.8%), and rural(19.4%) areas.

Survey Methodology and Demographics

A stakeholder survey was conducted to evaluate existing garbage collection systems and identify user expectations for the proposed CleanLand system. The survey was distributed via Google Forms during November–December 2025 and received responses from 36 participants representing urban (31.4%), suburban (49%), and rural (19.6%) areas.

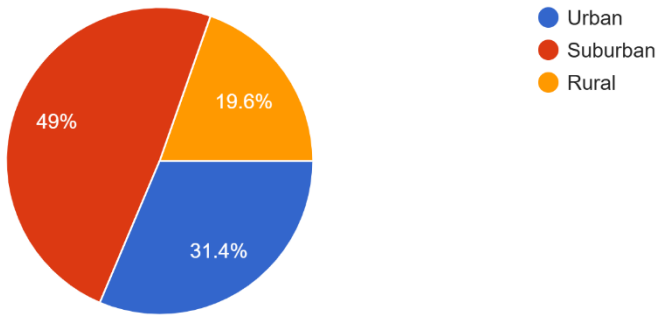


Figure 7:Geographical distribution of survey responders

Survey results indicates very low satisfaction with current garbage collection services. Only 2% of responders reports satisfaction, while 13.7% responders reports 75% satisfaction and 45.1 responders reports only 25% satisfaction. 13.7% responders reports 0% satisfaction, This shows widespread dissatisfaction caused by unreliable services, poor communication, and limited system transparency. Shows the need for improved garbage management system

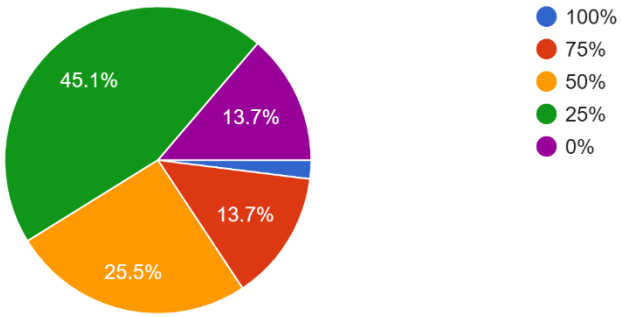


Figure 8: Satisfaction level with current systems

Notable communication gap was identified through the survey. 29.4% of responders says they were aware of garbage collection schedules, 70.6% says receiving no schedule related

information at all. This lack of effective communication leads to missed collections, inconvenience, and reduced service efficiency.

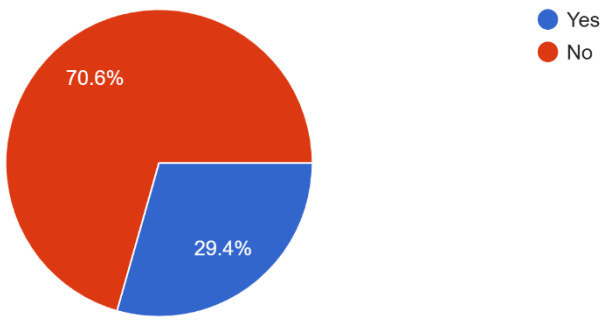


Figure 9: Schedule awareness among residents

The survey results indicate a significant lack of communication regarding garbage collection schedules. Out of 51 respondents, the majority (74.5%) reported that they do not receive any information at all about collection schedules. Only a small percentage receive updates through WhatsApp messages (11.8%), phone calls (7.8%), or by visiting official websites (3.9%), while none receive information via email. These findings highlight a major communication gap in the current garbage collection system and emphasize the need for a centralized digital platform .

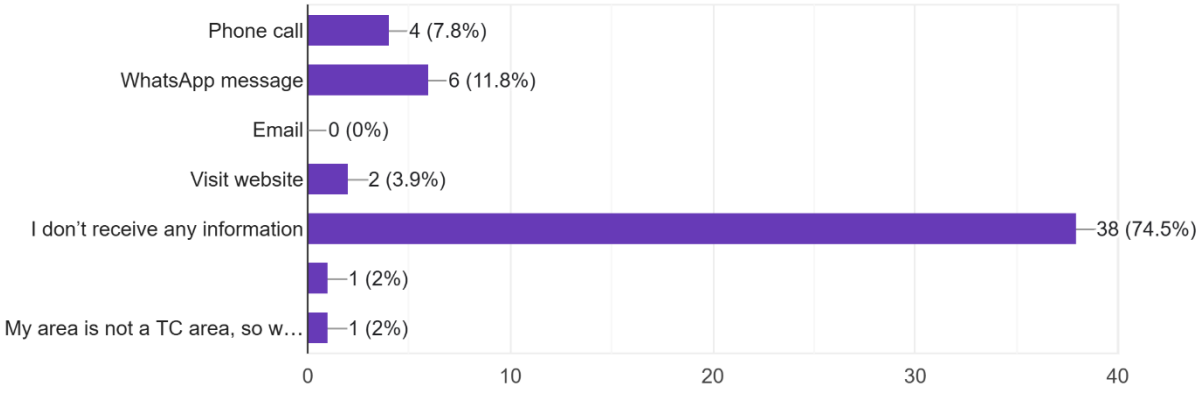


Figure 10: Information on garbage collection schedule

Survey reveled major reliability issues, with 86% responder experience missed garbage collections within the last 3 months. Such frequent failure raise environmental and public health concerns and highlight the accountability, and real-time operational control in existing systems.

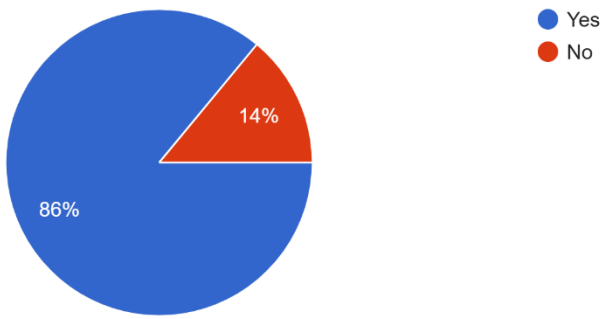


Figure 11: Frequency of missed garbage pickup

Complaint management also found to be ineffective. Although service issues are common, 62.7% responders do not report problems due to inconvenient or ineffective reporting methods. Those who do report issues rely on in person visits or phone call, indicating the lack of accessible digital complaints submission and tracking facilities

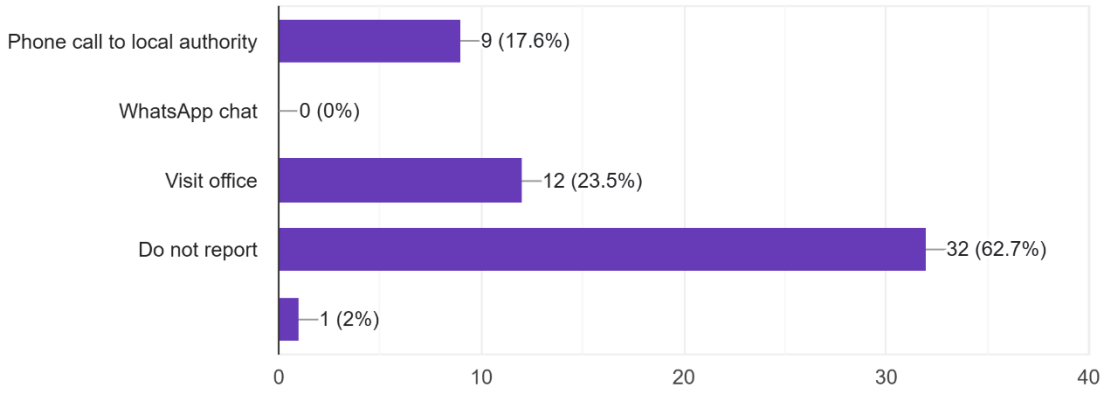


Figure 12: Complaint submission methods

Most respondents confirmed the absence of digital monitoring features. About 92.2% do not have access to real-time garbage truck tracking, and 7.8% do not receive bin or collection updates. This lack of visibility forces residents to depend on manual observation and increases uncertainty regarding service delivery.

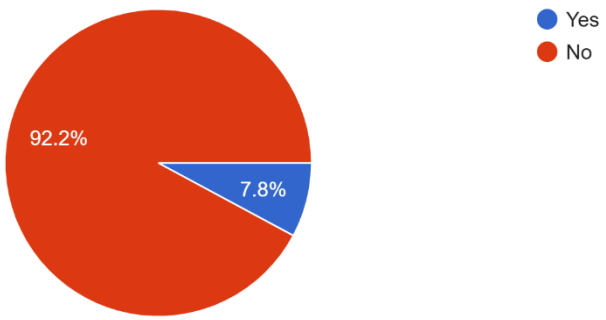


Figure 13: Access to truck tracking

94% stated that they do not receive any garbage bin updates, while only 6% confirmed receiving updates. Furthermore, none of the respondents reported receiving updates through modern digital methods such as QR code scanning. This highlights the absence of a structured and technology-driven bin monitoring system and emphasizes the need for features such as QR-based bin identification and real-time bin status updates in the proposed system.

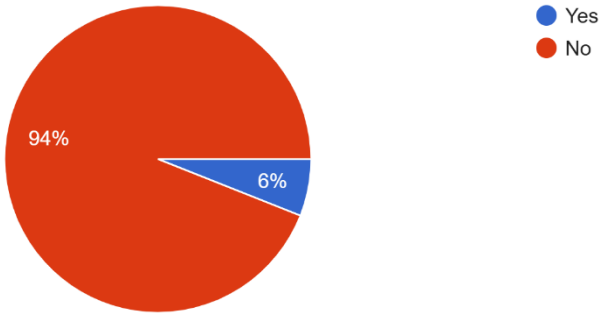


Figure 14: Availability of bin update

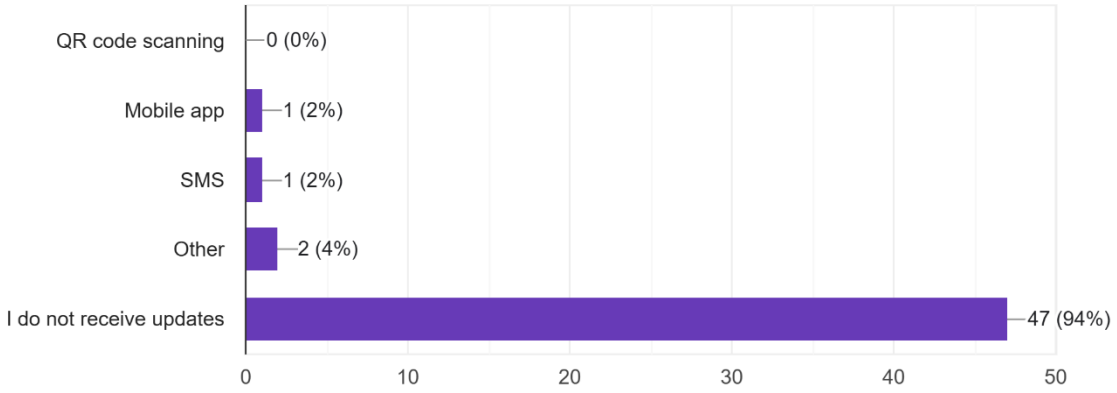


Figure 15: Garbage Bin Update Reception

Respondents identified key features required in a digital waste management system, including schedule viewing (58.8%), complaint submission with images (66.7%), complaint status tracking (56.9%), and automated notifications (43.1%). These priorities emphasize the demand for transparency, accountability, and user-friendly digital interaction.

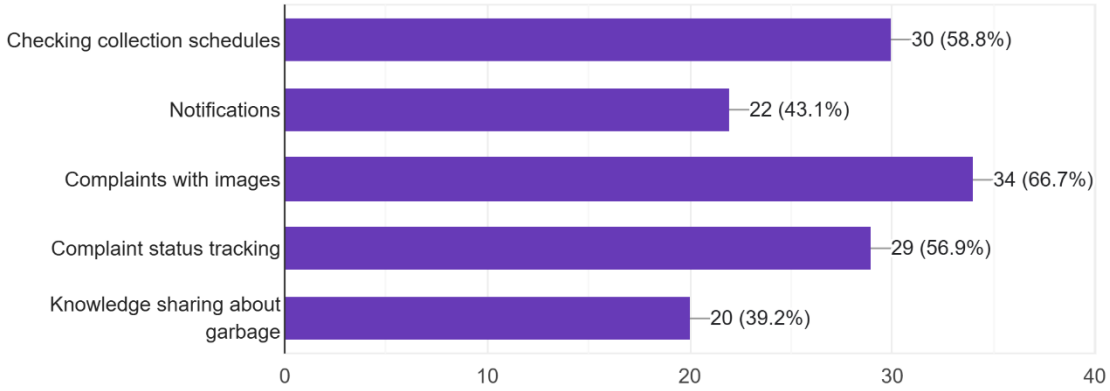


Figure 16: Stakeholder expected features

Overall, the survey confirms major shortcomings in existing garbage management systems, particularly in communication, reliability, and user engagement. The findings provide

strong justification for the CleanLand system, which aims to introduce real-time tracking, digital complaints, notifications, and QR-based bin management. Addressing these gaps will enable CleanLand to deliver a more transparent, efficient, and citizen-centric waste management solution.

3.Project Plan and Initial Design

This section present the methodology and the implementation plan for the CleanLand Garbage Management System. The proposed solution integrates software, database, and hardware components to deliver an efficient, web based garbage collection management system. The proposed methodology for this project is a waterfall model approach.

3.1. Components of the System

The component of the system includes software component, database component, and hardware component. Each component plays a crucial role in ensuring the effective functioning of the overall system. It is a web based application and database is used to store data.

3.1.1. Software Component and Tasks

The software component is the core part of the CleanLand Garbage Management System. It developed as a responsive web based application accessible through modern browsers on desktop and mobile devices. This system will facilitate user (residents, driver/cleaner) registration and & profile management, real time garbage truck tracking, garbage collection schedule view, Complaint management system, bin, route & driver assignment, awareness & information sharing, special pickups request management, QR code bin identification, Emergency alert, public feedback and rating system

Wireframes of key functionalities are listed below

Resident registration

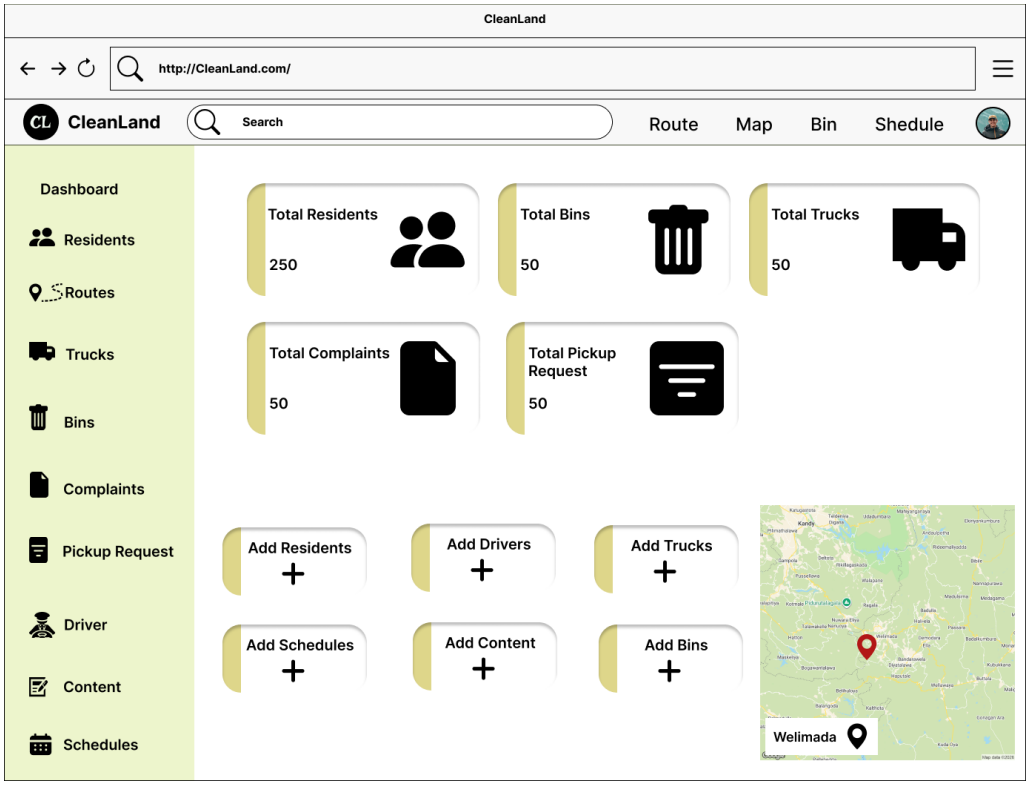


Figure 17: Admin dashboard UI

Complaint submission form

Residents can submit complaints regarding missed pickups, overflowing bins, or service issues. Admin can manage complaints

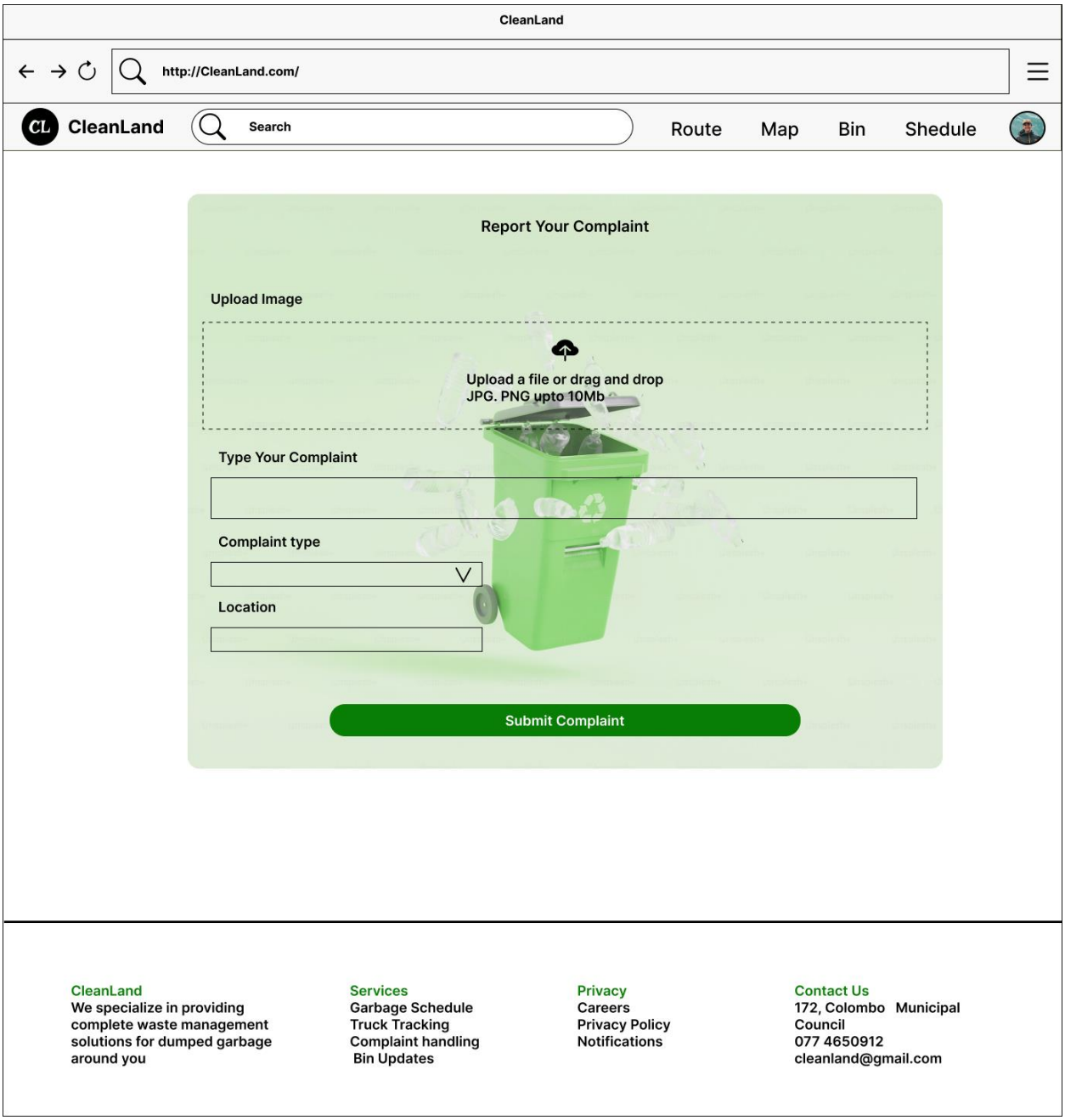


Figure 18: Complaint UI

Truck tracking page

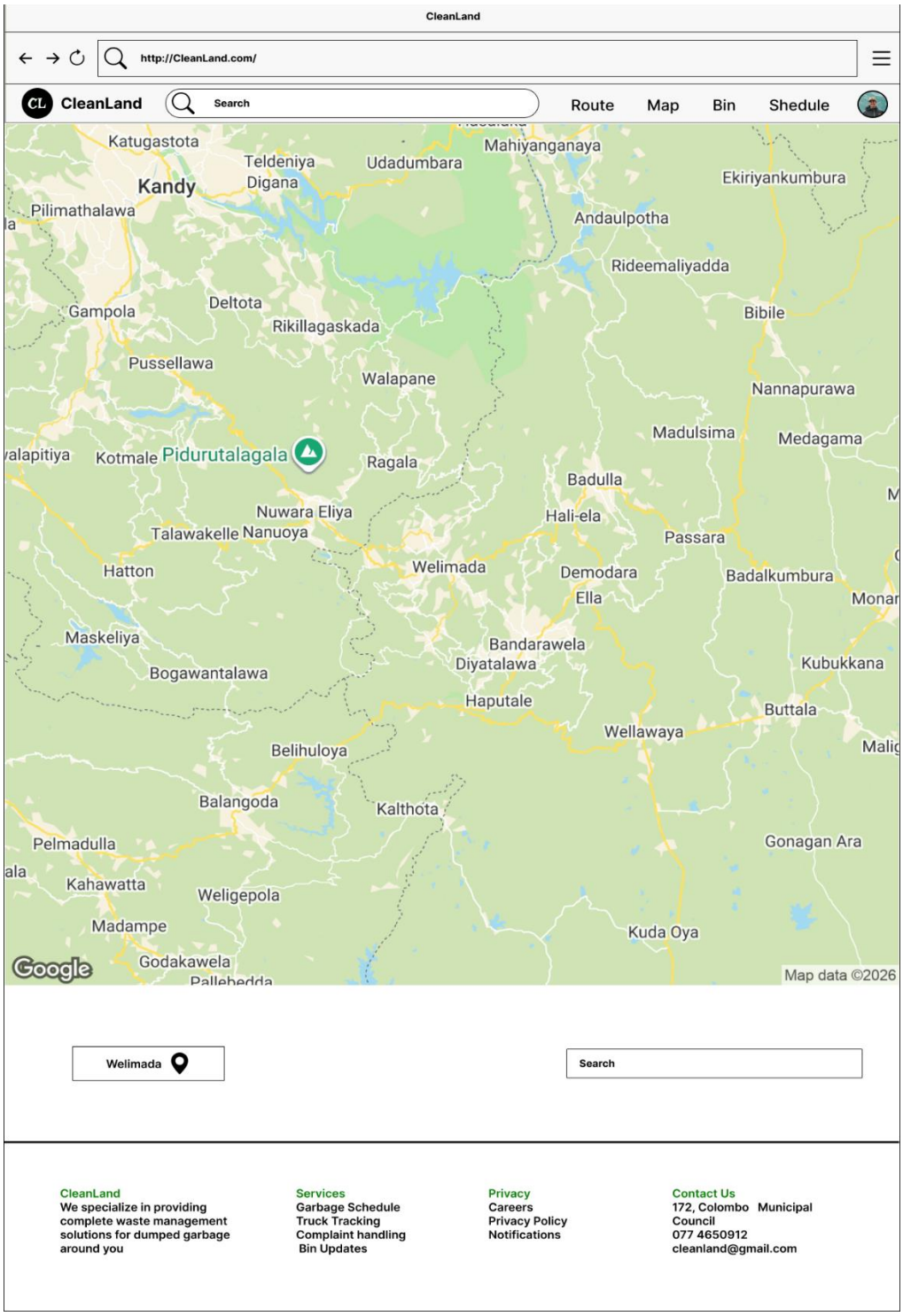


Figure 19: Truck tracking UI

Login form

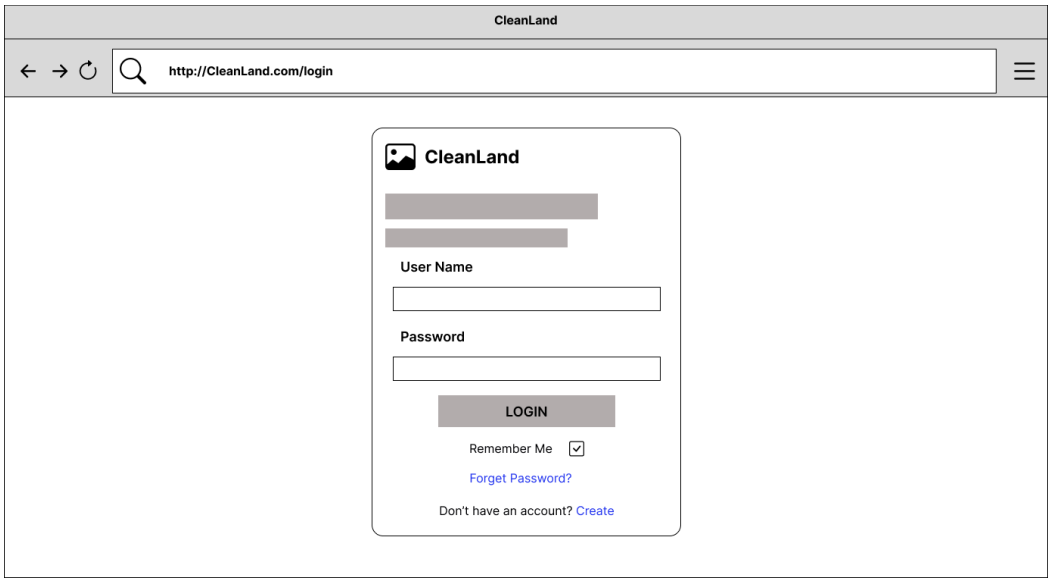


Figure 20: Login UI

Collection schedule

Admin can manage garbage collection schedules, and residents can view schedules based ion their area.

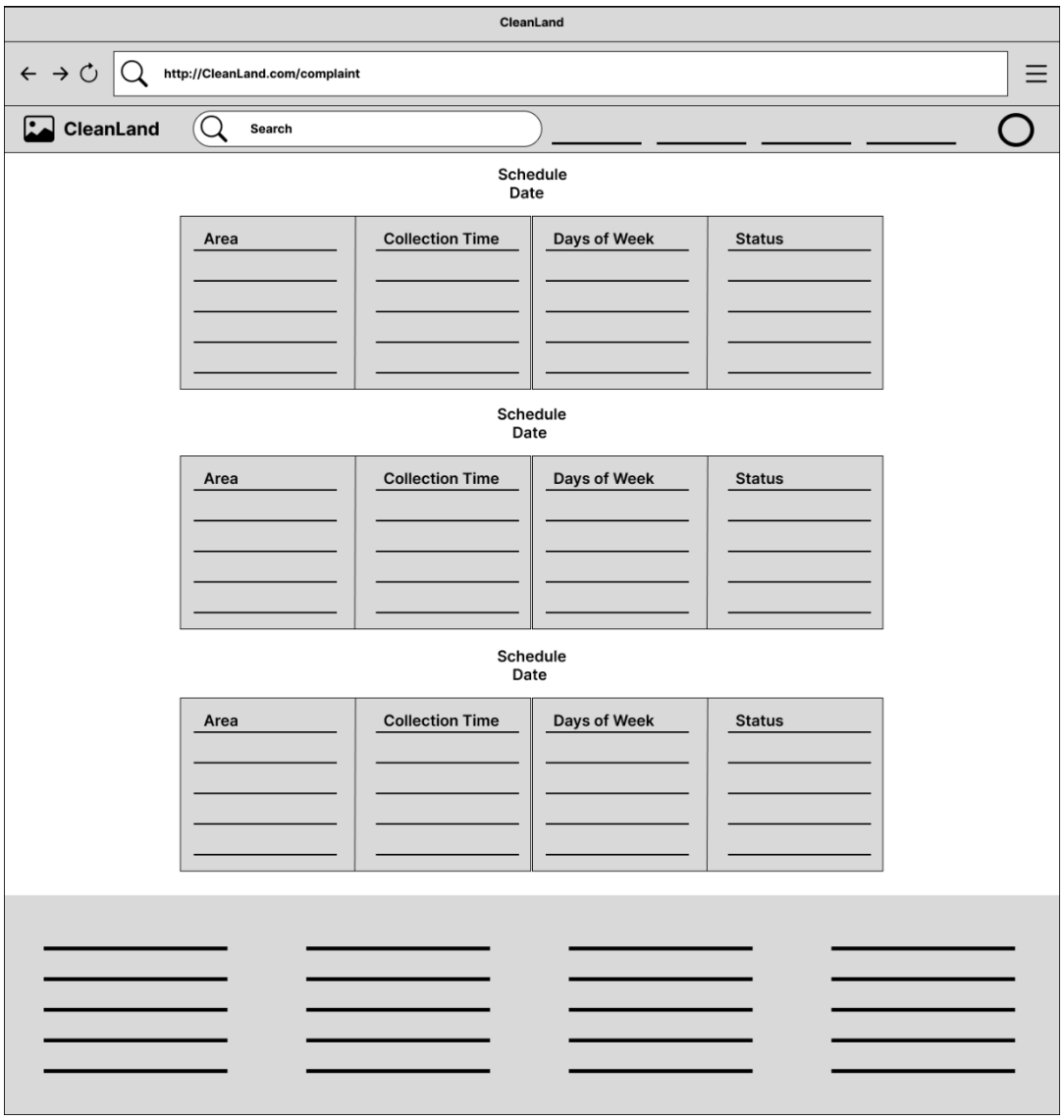


Figure 21: Collection schedule UI

Garbage truck tracking

This component enables real time tracking of garbage trucks using GPS data integrated with google maps. Residents and administrators can view live truck location

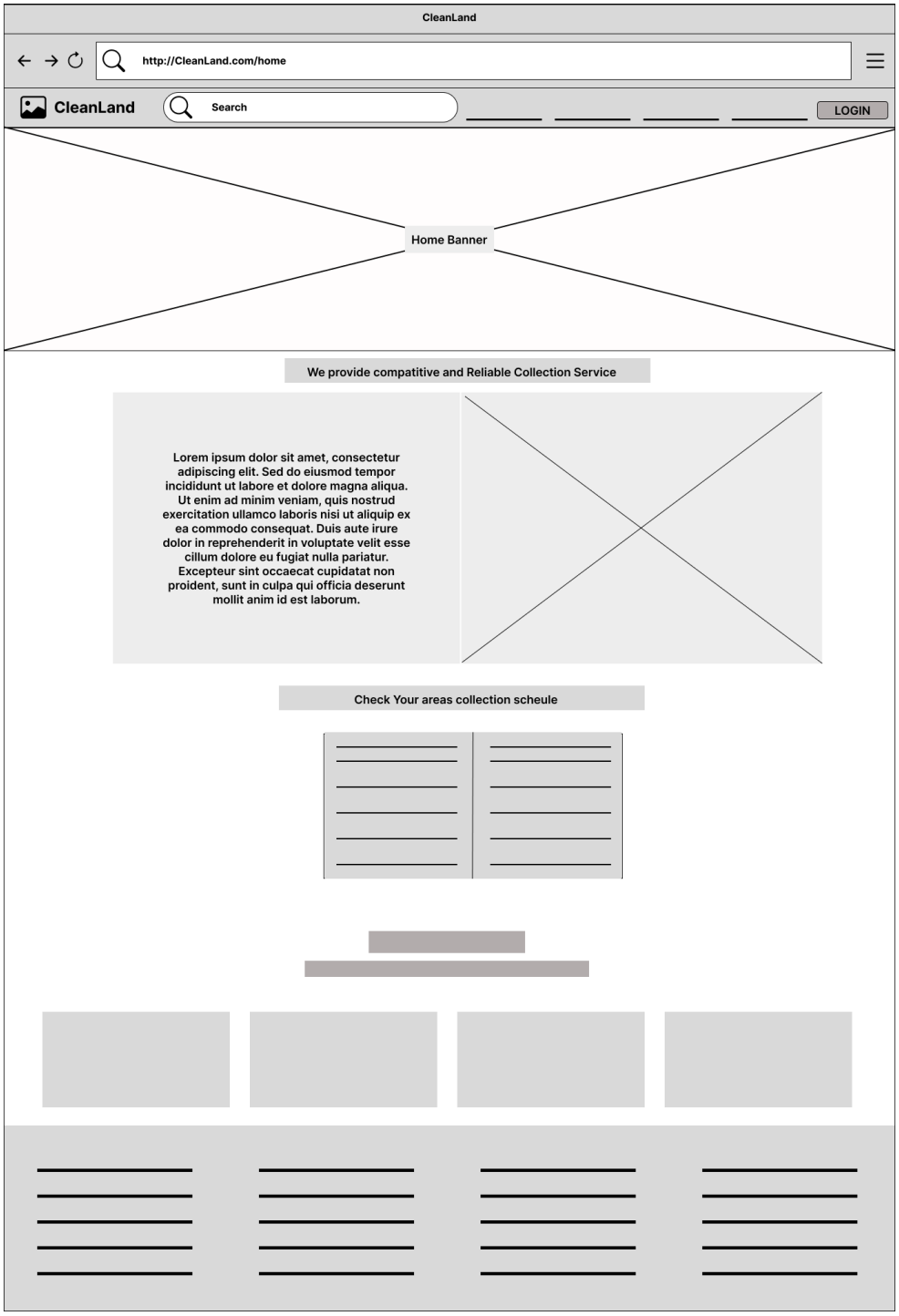


Figure 22: Garbage truck tracking UI

Feedback component

Residents can submit feedback and rate services and website

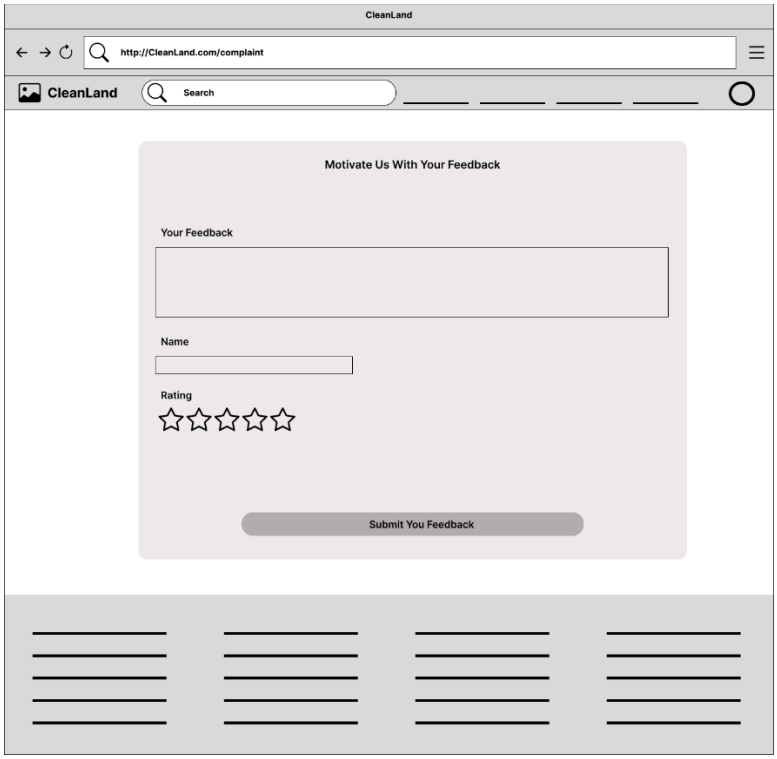


Figure 23: Feedback submission UI

QR code bin scanning component

Each garbage bin is has a unique QR code. Residents and drivers can scan the QR code to view bin details and update collection status.

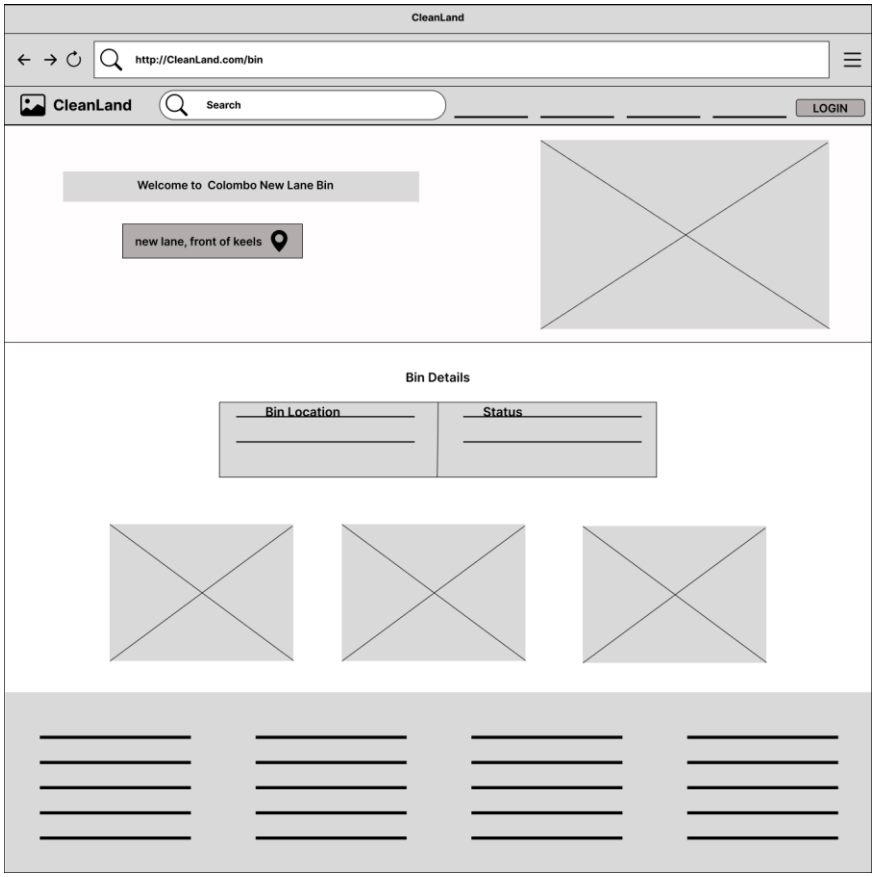


Figure 24: QR code bin scanning UI

User dashboard

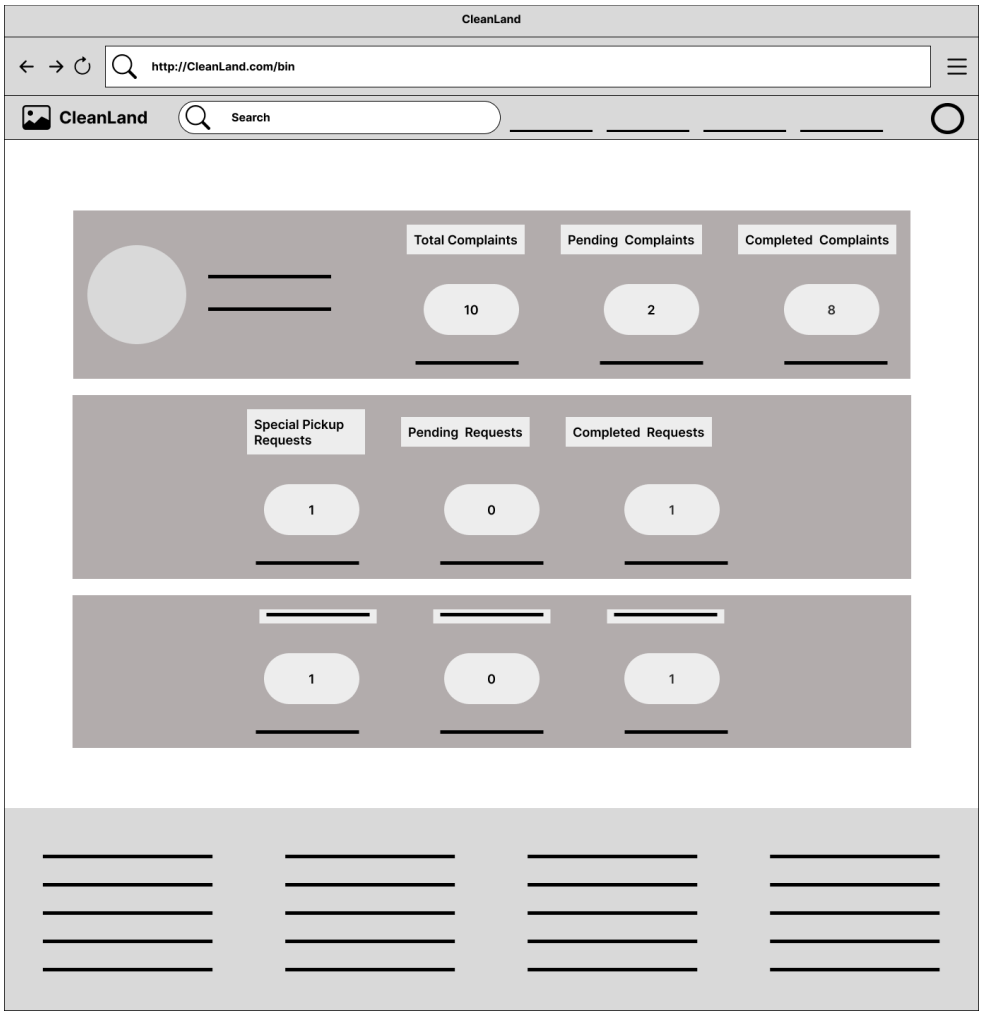


Figure 25: Resident dashboard UI

Registration form

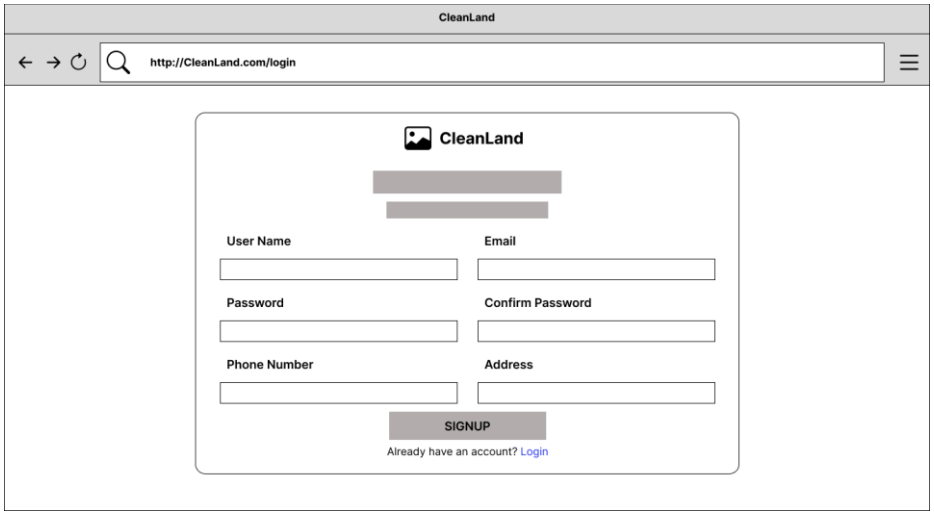


Figure 26: Registration UI

Admin content upload component

Admin uploads contents here regarding recycling tips, health guidelines, and updates on environment events**.**



Figure 27: Content upload UI

3.1.2. Database Component

This chapter presents the entities identified in the project along with their attributes and associated ERD. This is a fundamental requirement for the overall functionality of the system.

Driver table

Attribute	Data Type	Description
Driver_id	Varchar (6)	Primary key, unique identifier for the driver
Email	Varchar (255)	Email of the driver
Name	Varchar (255)	Name of the driver
Password	Varchar (25)	Password of the account
Phone_no	Varchar (20)	Contact number of driver
Truck_id	Varchar (6)	Foreign key referring_ truck table

Table 8: Driver table

Resident table

Attribute	Data Type	Description
Resident_id	Varchar (6)	Primary key, unique identifier for the resident
Name	Varchar (255)	Full name of resident
Password	Varchar (255)	Hashed password for account security
Address	Varchar (255)	Residential address of the address
Phone_no	Varchar(20)	Contact number of the resident
Email	Varchar (255)	Email of the resident

Table 9: Resident table

Complaint table

Attribute	Data Type	Description
Complaint_id	Varchar (6)	Primary key, unique identifier for the complaint
Complaint_type	Enum	Category of the complaint
Description	Text	Detailed description of the complaint
Image	Varchar	File path or URL to complaint image
Location	Varcha(255)	Location where the complaint occurred
Status	Enum	Current status of the complaint

Resident_id	Varchar (6)	Foreign key referencing resident table

Table 10: Complaint table

Bin table

Attribute	Data Type	Description
Bin_id(PK)	Varchar (6)	Unique identifier for the bin table
Route_id (FK)	Varchar (6)	Foreign key referencing Route
Current_status	Enum	Current fill status of the bin
Bin_location	Varchar (6)	Location of the bin
Bin_type	Enum	Type of waste the bin collects
QR_id (FK)	Varchar (6)	Foreign key referencing QR code table

Table 11: Bin table

Truck table

Attribute	Data Type	Description
Truck_id (PK)	Varchar (6)	Unique identifier for the truck table
Driver_id (FK)	Varchar (6)	Foreign key referencing driver table
Vehicle_no	Varchar (20)	Registration/ license plate number

Table 12: Truck table

Schedule table

Attribute	Data Type	Description
Schedule_id	Varchar (6)	Unique identifier for the schedule
Status	Varchar(50)	Status of the scheduled collection
Area	Varchar (255)	Geographic area/neighborhood for collection
Collection_time	Time	Scheduled time for garbage collection
Day_of_week	Enum	Day when collection is scheduled
Route_id	Varchar (6)	Foreign key referencing Route table

Table 13: Schedule table

Feedback table

Attribute	Data Type	Description
Feedback_id (PK)	Varchar (6)	Unique identifier for the feedback
Resident_name	Varchar (255)	Name of the resident providing feedback
Comments	Varchar (255)	Written feedback/comments from the resident
Rating	Int	Rating score (1-5)
Submitted_date	Date	Date when feedback was submitted
Resident_id	Varchar (6)	Foreign key referencing Resident table

Table 14: Feedback table

GPS tracker table

Attribute	Data Type	Description
Tracker_id	Varchar (6)	Primary key, unique identifier for GPS tracking entry
Timestamp	DateTime	Date and time of the GPS reading
Current_status	Enum	Current activity status of the truck
Latitude	Decimal	Geographic latitude
Longtitude	Decimal	Geographic longitude
Status	Varchar(50)	Status of the truck
Speed	Decimal	Current speed in km/h
Truk_id	Varchar (6)	Foreign key referencing Truck table

Table 15: GPS tracker table

Awareness content table

Attribute	Data Type	Description
Content_id	Varchar (6)	unique identifier for the content
Title	Varchar (255)	Title of the awareness content
Content_type	Enum	Type/category of the content
Image	Varchar (255)	File path or URL to associated image

Content_body	Text	Main body/text of the content
Description	Text	description of the content

Table 16: Awareness content table

Emergency alert table

Attribute	Data Type	Description
Alert_id (PK)	Varchar (6)	Unique identifier for the alert
Alert_type	Enum	Type/category of the emergency alert
Title	Varchar (255)	Alert title/headline
Message	Text	Detailed alert message
Create_time	DateTime	Date and time when alert was created
Expire_time	DateTime	Date and time when alert expires/ends
Is_active	Boolean	Flag indicating if alert is currently active

Table 17: Emergency alert table

Notification table

Attribute	Data Type	Description
Notification_id	Varchar (6)	Primary key, unique identifier for the notification
Notification_type	Enum	Type/category of notification
Title	Varchar (255)	Notification title/subject
Message	Text	Notification message content
Sent_time	DateTime	Date and time when notification was sent
resident_id	Varchar (6)	Foreign key referencing Resident table

Table 18: Notification table

Route table

Attribute	Data Type	Description
Route_id	Varchar (6)	Unique identifier for the route

Route_name	Varchar (255)	Name/identifier of the route
Area_name	Varchar (255)	Name of the area/neighborhood covered
Strat_location	Varchar (255)	Starting point of the collection
End_location	Varchar (255)	Ending point of the collection route
Estimated_duration	Time	durationTimeExpected time to complete the route
Driver_id	Varchar (6)	Foreign key referencing Driver table
Truck_id	Varchar (6)	Foreign key referencing Truck table

Table 19: Route table

Special request table

Attribute	Data Type	Description
Request_id	Varchar (6)	Primary key, unique identifier for the request
Request_time	Time	Time when request was submitted
Request_date	Date	Date when request was submitted
Location	Varchar	Pickup location for the special request
Email	Varchar (255)	Contact email of the requester
Description	Text	Detailed description of items/ request
Address	Varchar (255)	Full address for pickup
Status	Varchar(50)	Current status (e.g., Pending, Approved, Scheduled, Completed, Rejected)
Scheduled_date	Date	dateDateDate when pickup is scheduled
Item_type	Enum	'Furniture', 'Electronics', 'Construction Waste', 'Festival Waste', 'Bulk Items', 'Other')Type of items for special pickup

Driver_id	Varchar (6)	Foreign key referencing Driver table
Scheduled_time	Time	Scheduled time for pickup

Table 20: Special request table

QR code table

Attribute	Data Type	Description
QR_id	Varchar (6)	Primary key, unique identifier for the QR code
Bin_id	Varchar (6)	Foreign key referencing Bin table
Data	Text	Encoded QR code data string

Table 21: QR code table

3.2. Proposed Methodology

The CleanLand Garbage Management system will be developed using the waterfall development methodology. The waterfall model follows a structured and sequential approach, where each phase of development is completed before moving on the next

The waterfall method was determined for this project due to below reasons

- The project requires approval from the university in several phases before moving to the next phase.
- Ability to track the progress across 6 months
- Well defined deliverables at each stage
- The structured nature of the Waterfall model makes it easier to manage timelines, responsibilities, and assessment milestones within an academic project environment.

3.3. Technologies Adapted

Frontend- React.js

React.js makes it possible to create responsive interfaces, interactive and user friendly interfaces. React's component based architecture enables reuseable user interface components, which reduces development time and enhance maintainability

Backend- Node.js

Node.js is selected for backend development, because its high in performance and non blocking architecture, which efficiently handles multiple user requests simultaneously.

The garbage management system involves real time data processing, such as GPS updates, notifications, and user interactions. Node.js enables fast server side execution and seamless integration with frontend technology React.js.

Database- MySQL

MySQL is used as the database management system because it provides a structured, reliable, and secure relational database environment, The system manages large amount of data including user profiles, schedules, complaints, routes, and bin information. MySQL ensures data consistency through relational constraints and supports efficient querying, which is essential for generating tracking complaint history and managing schedules.

Mapping- Google Maps

Google Maps API is used to implement real-time garbage truck tracking and location visualization. It provides accurate mapping, GPS integration, and route visualization features that are critical for displaying truck locations and estimated arrival times. The API is widely supported, reliable, and easy to integrate with web applications, making it suitable for both administrators and residents to monitor garbage collection activities effectively.

3.4. Test and Deployment Plan

The CleanLand garbage management system will be tested in all aspects to ensure it is suitable for deployment.

- Unit testing- Each individual unit will be tested individually using tools such as Jest or Mocha.
- Integration testing- Testing will be done to make sure that the units are working together correctly using tools such as Post Man or Jest.
- Functionality testing- Upon integration each functional requirement will be tested.
- UI testing- Usability and user friendly of the system will be tested here.
- Regression testing- The functioning of the system when code fixes are applied will be tested here.
- Performance testing- Simulates high user condition using JMeter and test the performance of the testing.

- UAT- Conduct user acceptance testing with stakeholders and getting feedback on the system.

The system will be deployed on a web server accessible to municipal council. The deployment will be take place in step by step such as deployment of backend and database, deployment of the web site and performing live testing.

4.References

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5.Appendix

5.1. Timeline - Gantt chart

| 2025-2026 | | |

5.2. Use Case Diagrams

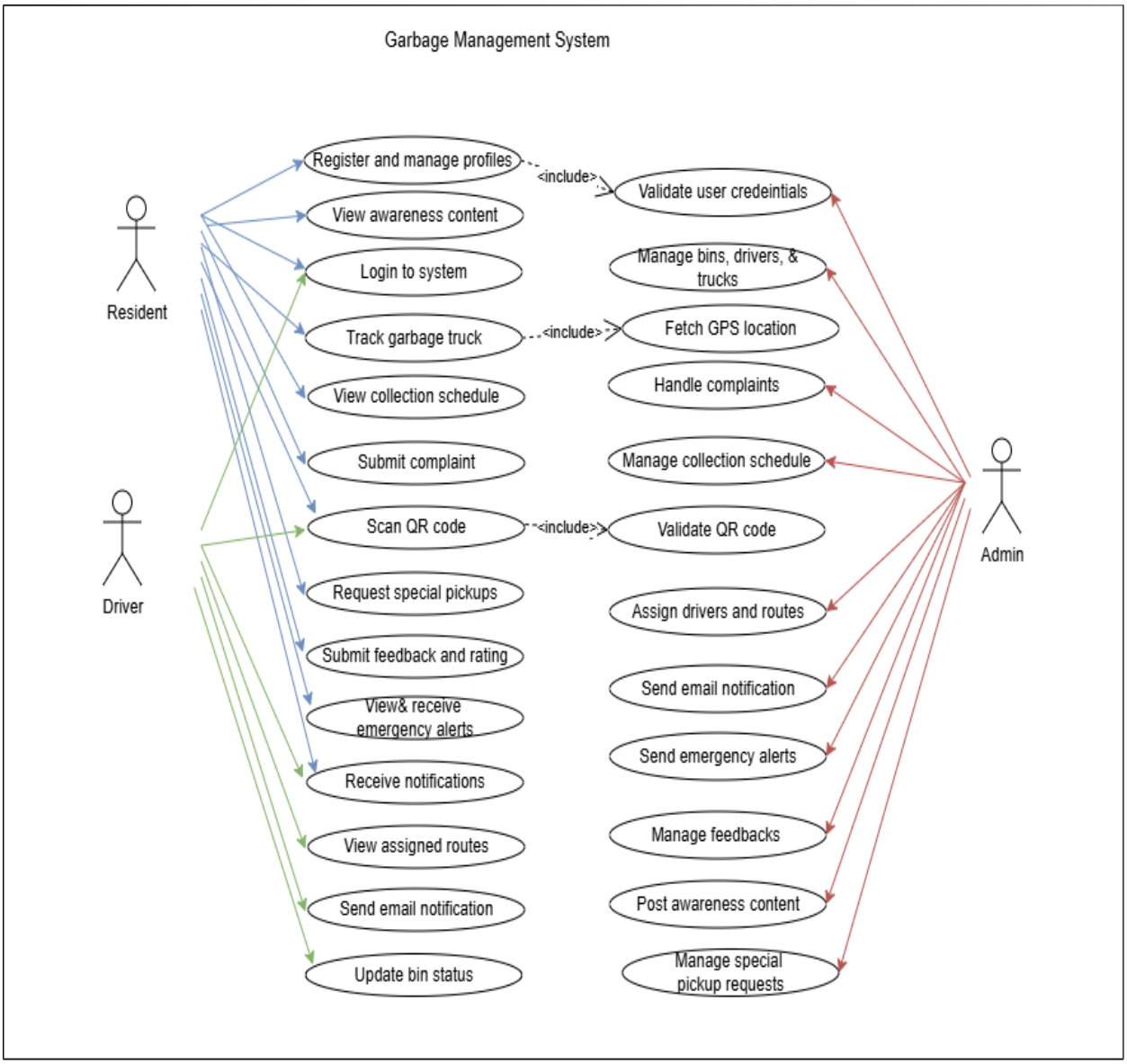


Figure 28: Use case diagram

5.3. Activity Diagrams

Complaint Management

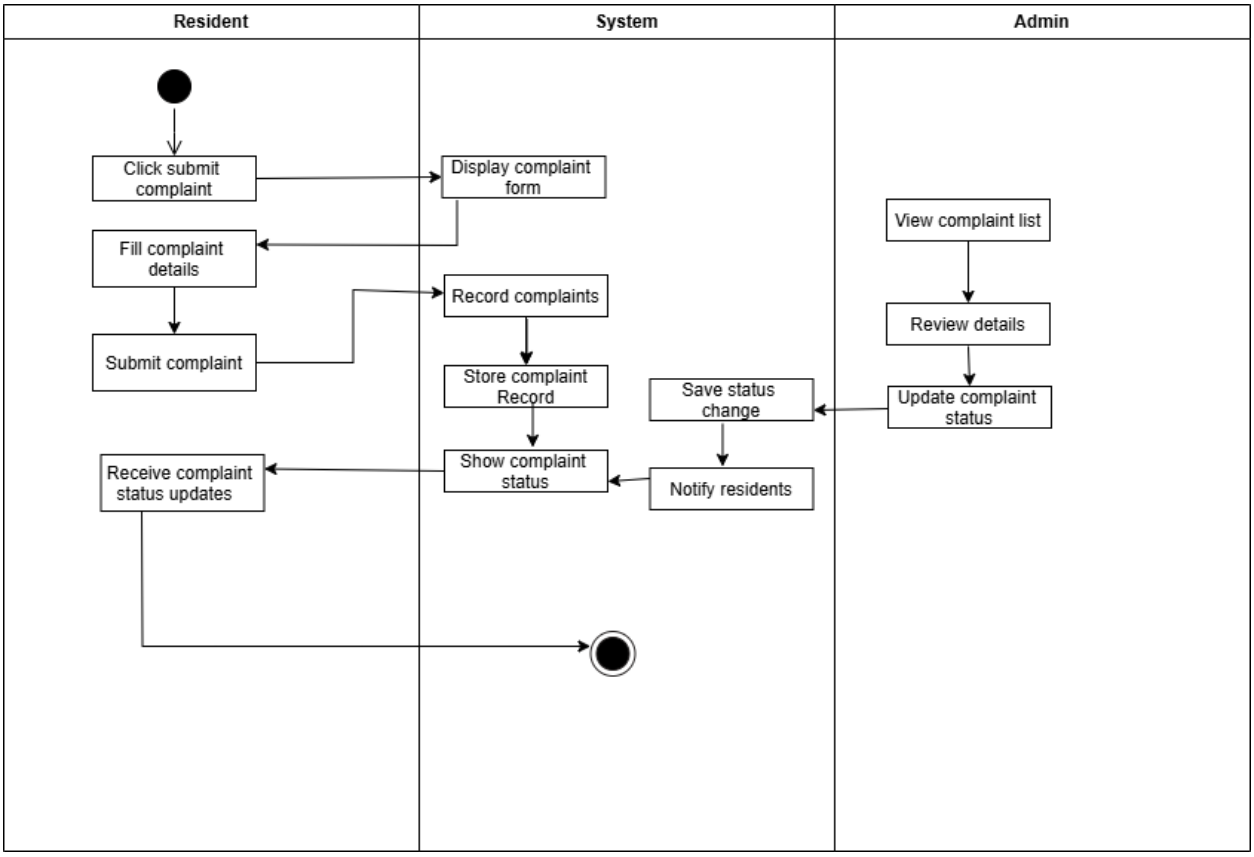


Figure 29:Complaint management diagram

Garbage Truck Tracking

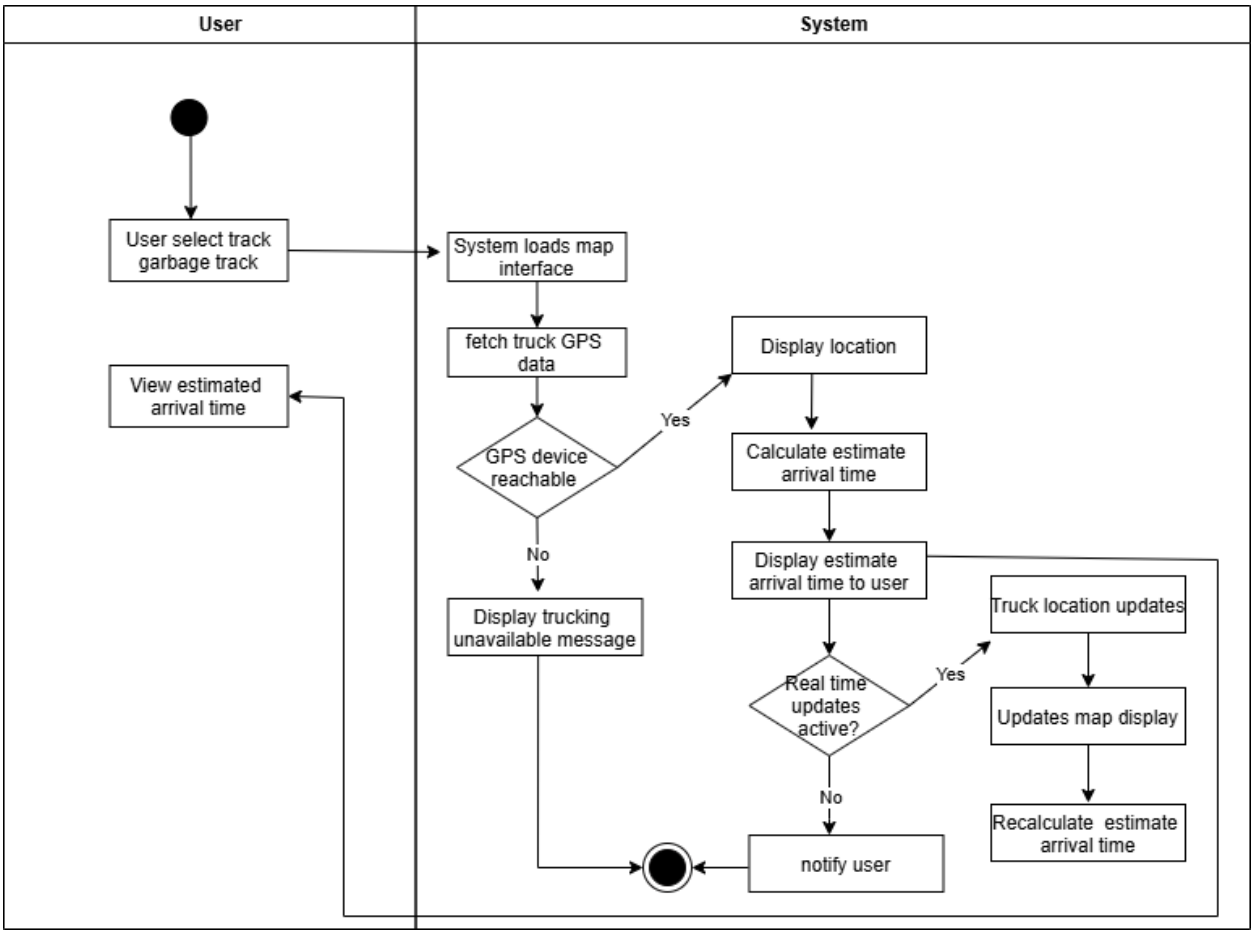


Figure 30:Garbage truck tracking activity diagram

QR Code Bin Identification

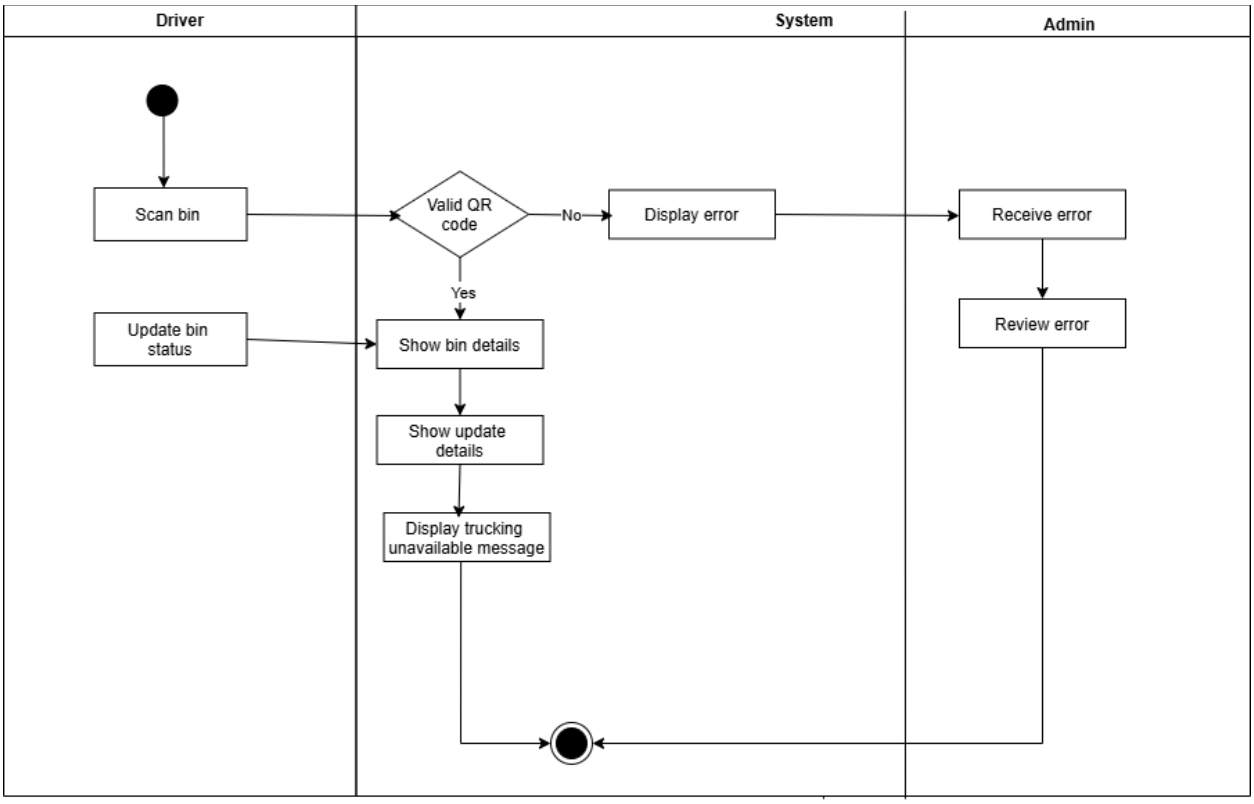


Figure 31: QR code bin identification

Resident Registration

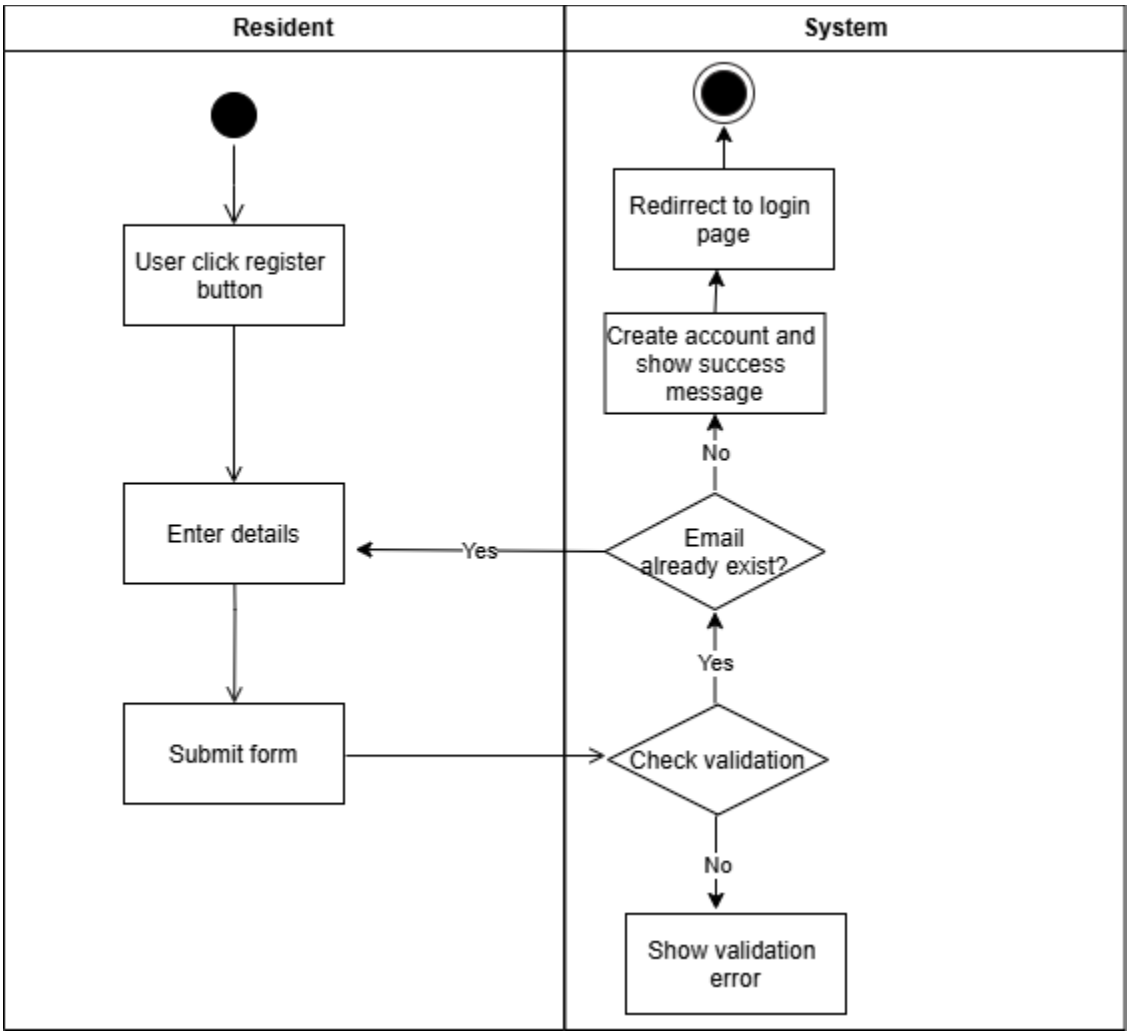


Figure 32: Resident registration activity diagram

Login to the System

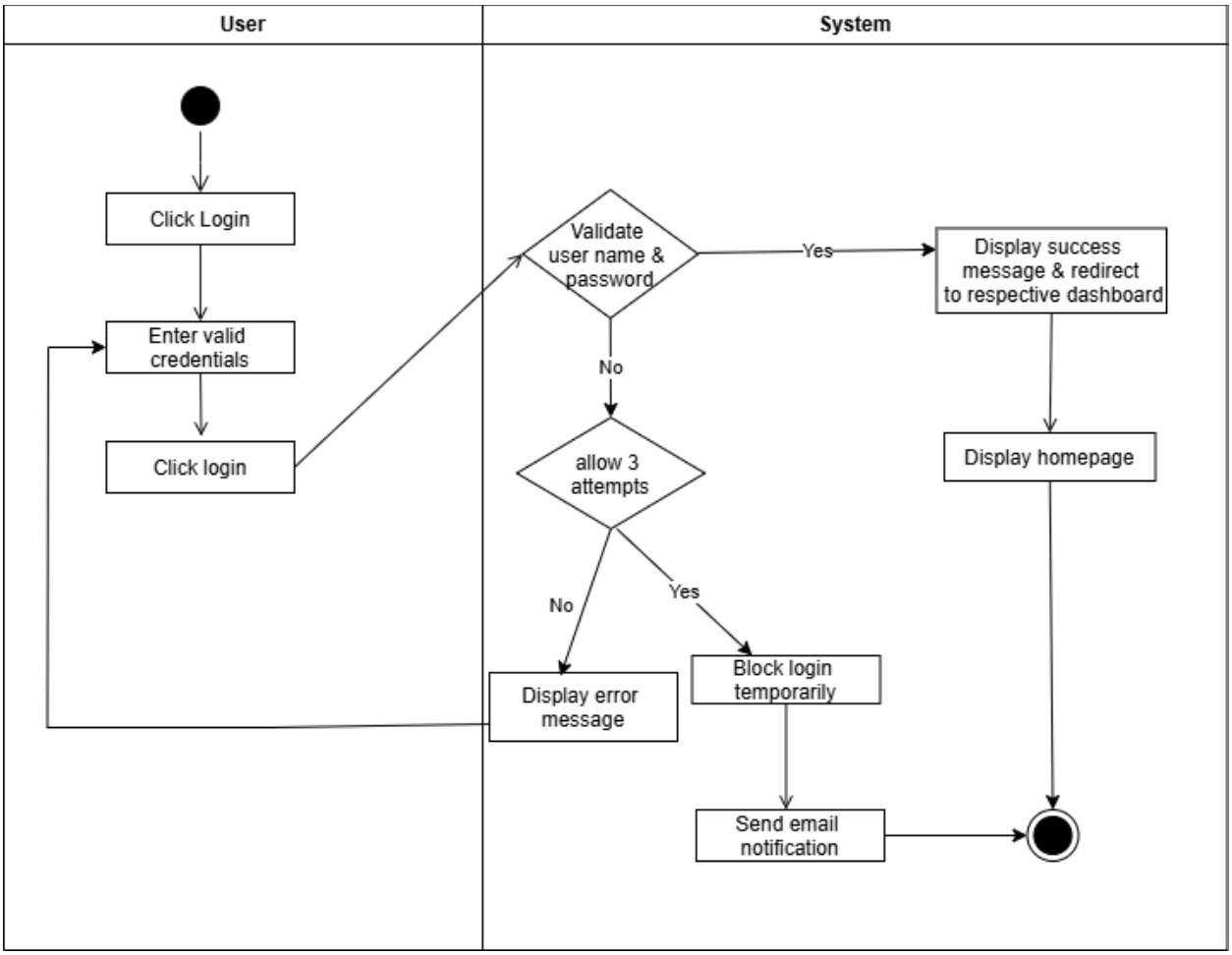


Figure 33: System login activity diagram

Garbage Collection Scheduling

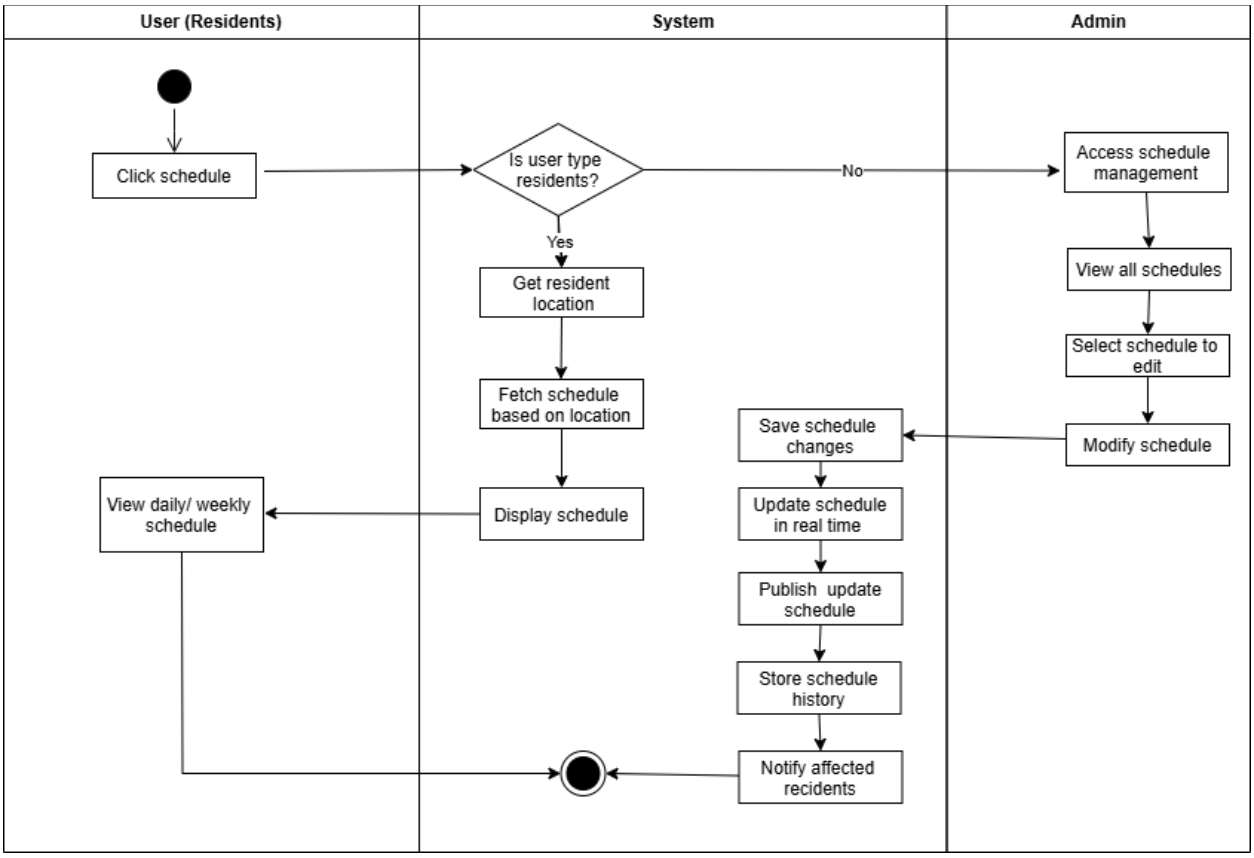


Figure 34: Garbage collection schedule activity diagram

Emergency Alerts

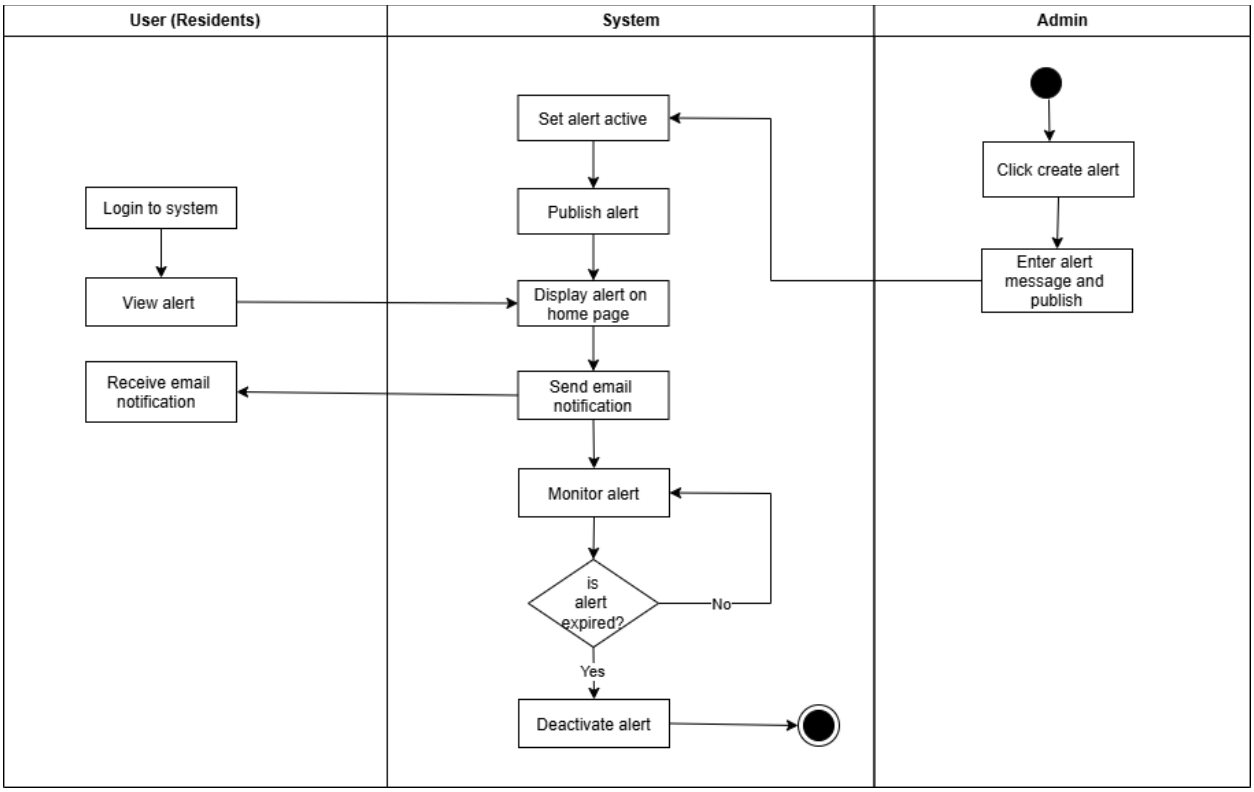


Figure 35: Emergency alert activity diagram

Public Feedback and Rating System

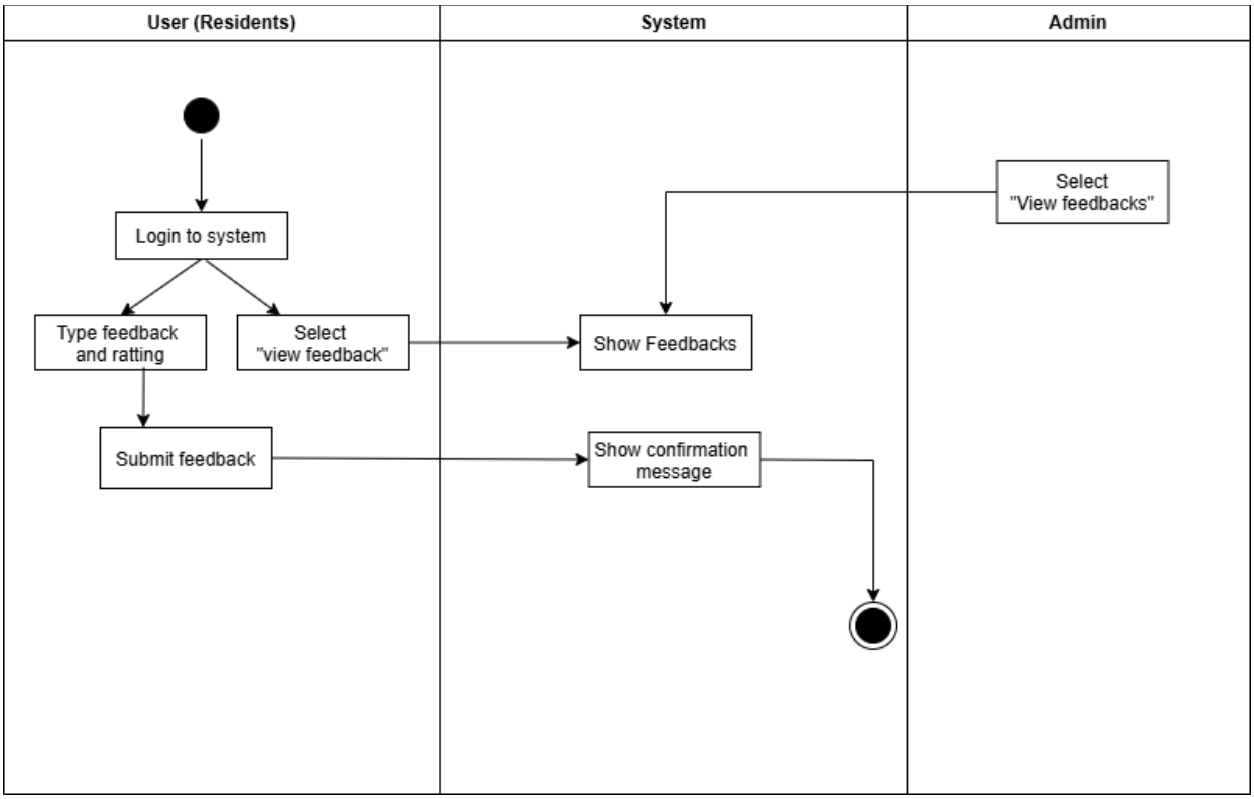


Figure 36: Public feedback & rating activity diagram

Special Pickup Request

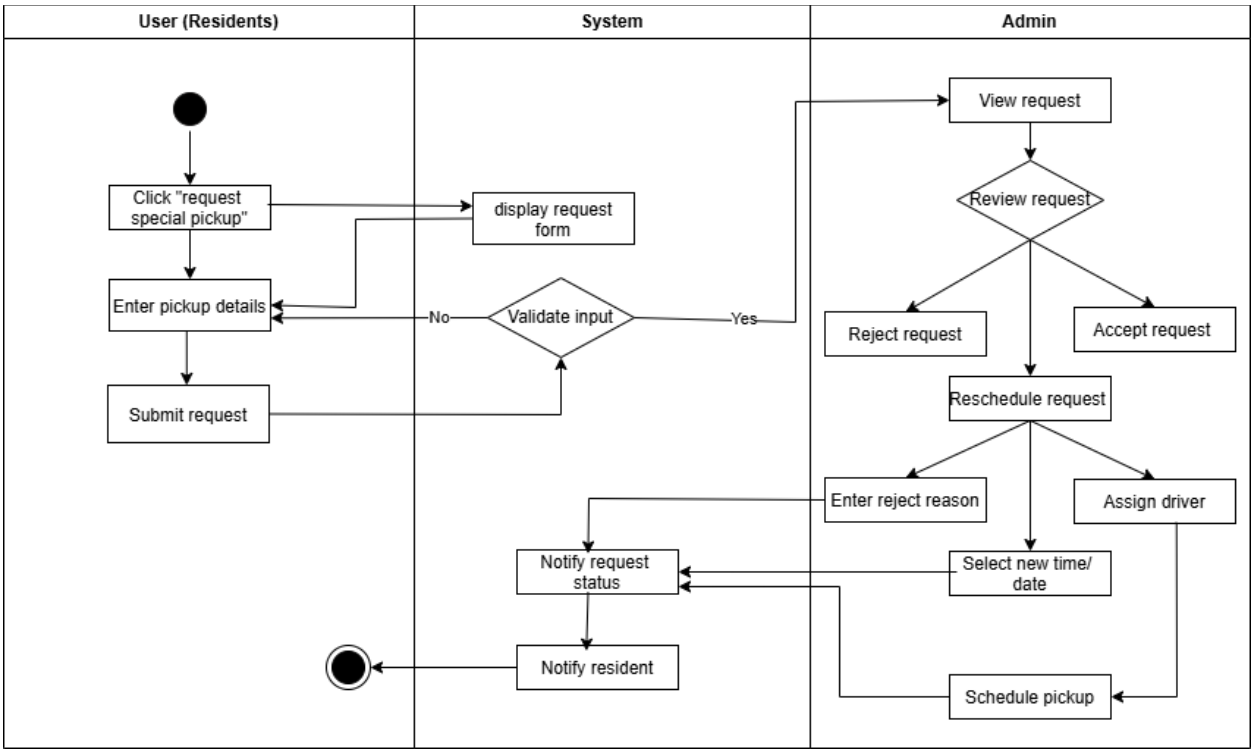


Figure 37: Special pickup request activity diagram

Public Awareness & Notifications

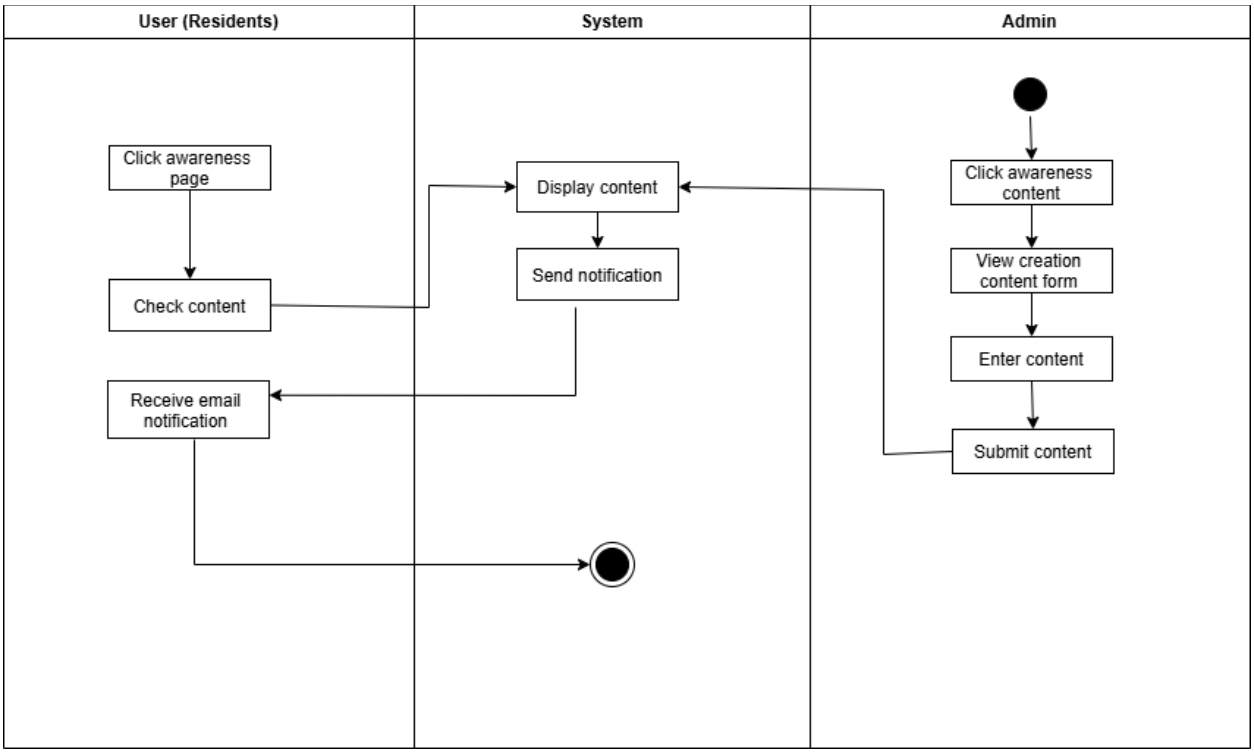


Figure 38: public awareness and notification activity diagram

5.4. ERD

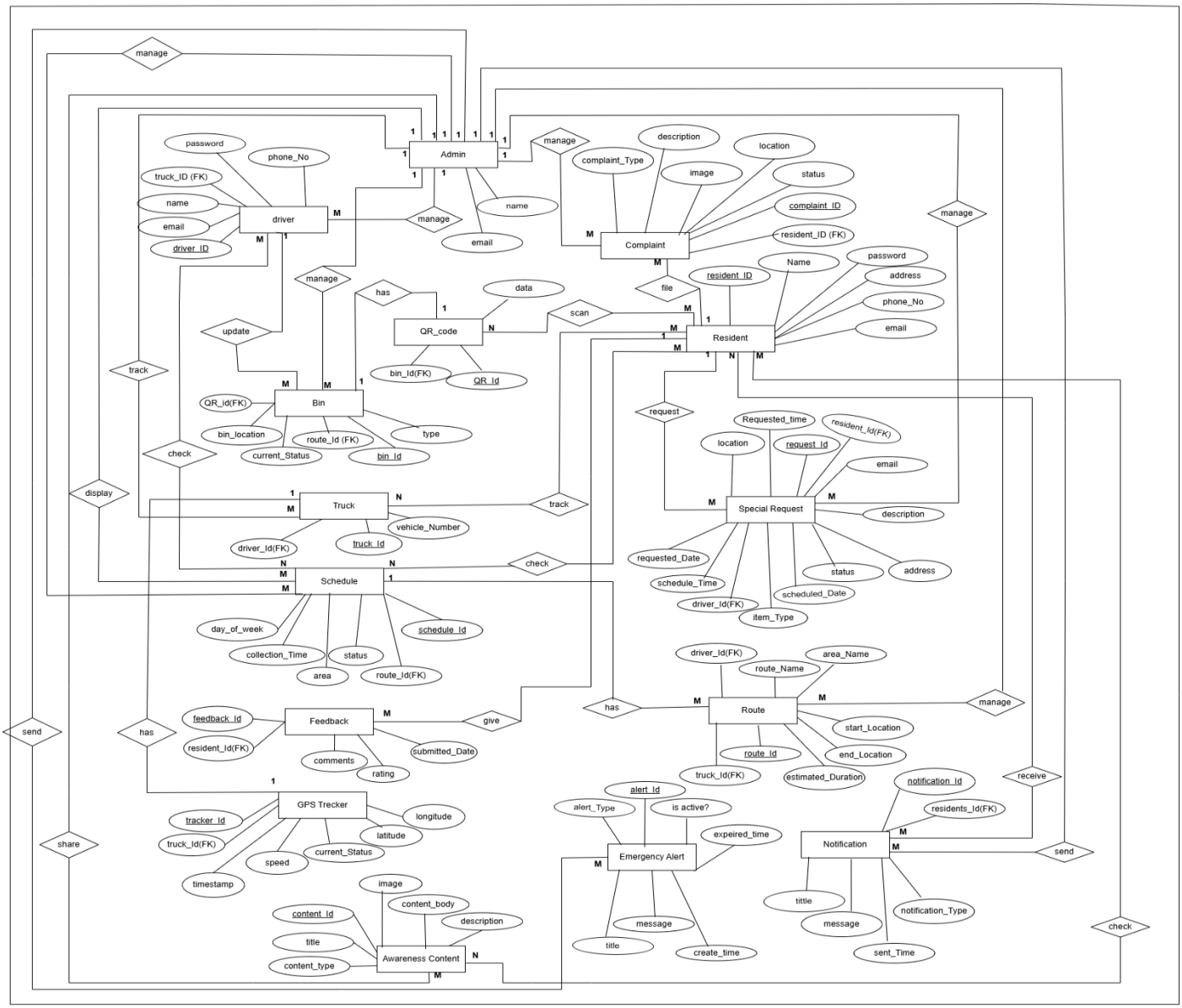


Figure 39: ER diagram

5.5 SRS

Software Requirements Specification for Garbage Management System

Version 1.0 approved

Prepared by E2447027

University of Moratuwa

11/26/2025

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of the system that is being proposed is to develop garbage collection management system, version 1.0. The garbage collection management system is web based platform developed to streamline the efficient garbage collection process by enhancing communication and coordination among residents, garbage collectors, and municipal councils and promote eco friendly waste disposal practice

This document defines the functional and non functional requirements of the entire system, including user registration, real time vehicle tracking, garbage bin identification, complaint management, scheduling, and administrative operations. The system aims to eliminate every single one of the problems that were caused by the manual garbage collection management and help the admins and residents to do a better garbage management.

1.2 Document Conventions

Category	Description
Font	Times New Roman
Size	12
Line & Paragraph Style	1.15
Style	Normal
Color	Black

Table 1: Document conventions

1.3 Intended Audience and Reading Suggestions

The target audience of this document is who are involve in garbage management in government service sector. The municipal council officers, garbage collectors, garbage truck drivers, and resident around the country will find this document useful in understanding how they can be a part of this productive garbage collection management and to create a beautiful and clean nation. Each and every human living around municipal councils will be the main target audience who would want to dispose garbage in proper way. This document will explain how they can use and navigate through the system to organize and manage their garbage efficiently.

1.4 Project Scope

The garbage management system is a web based platform developed to improve the efficiency, transparency, and sustainability of garbage collection services Currently garbage collection in many areas lacks proper scheduling and communication between municipal council and residents. Residents are often unaware of collection time, residents don't have knowledge about garbage bins types, and councils have limited tools to monitor collection progress and manage complaints.

This proposed system addresses these challenges by providing a centralized platform for all stakeholder

The detailed scope of the system is as follows:

Garbage management system is a critical and continuously evolving field that plays a major role in maintaining environment cleanliness and public health. It involves planning and coordinating the collection, transportation of garbage while minimizing environmental and health impacts.

At present, most garbage collection activities in Sri Lanka are managed manually. Municipal council often rely on outdated methods for recording and monitoring garbage truck movements, while residents remain unaware of the collection schedules. Complaints regarding missed pickups, overflowing bins, or irregular service must be filled in person, leading to delays and inefficiency. Data related to routes, drivers, and complaints are typically handled using paper based systems or basic tools such as spreadsheets, resulting in data loss, human error, and lack of transparency.

The aim of developing this proposed web based garbage collection management systems is to create an integrated digital platform that connects residents garbage collectors, and municipal councils. The system will automate the garbage collection process, enhance communication between stockholders, and promote sustainable waste management practices.

The detailed scope of the proposed system is as follows:

Admins/ Municipal council officers

- Register and create account for admin
- Track the garbage trucks
- Assign drivers, vehicles, & create bins

- Replay to complaints or take actions
- Conduct awareness campaign and post announcement or events for residents
- Send email notifications
- Manage special pickups
- Manage feedbacks and ratings
- Send emergency alerts through email
- Manage bins

Garbage truck drivers/ collectors

- Create profile and log into system
- Update bin collection routes and schedules
- Report operational issues
- Update complaints

Residents

- Create profile and log into system
- Scan the QR code and identify the bin
- Track the truck in real time and view collection schedule
- File a complaint & follow it
- View past complaint status
- Receive notifications via email
- Request special pickups
- Give ratings and feedbacks
- Receive emergency alerts via email

1.5 References

[1] https://www.colombo.mc.gov.lk/garbage-collection.php- Colombo Municipal Council Website

2. Overall Description

2.1 Product Perspective

The garbage collection management system is a new, self contained web based application design to replace the existing manual garbage collection management process used by many municipal councils. Currently, waste collection and monitoring rely heavily on physical records, phone calls, and direct communication between residents and municipal staff. This traditional approach leads to inefficient, delayed pickups, miscommunication, and lack of transparency.

The proposed garbage collection management system provides a centralized digital platform that integrates the key stakeholders, residents, garbage collectors, and municipal council administrators into one unified system. It is not a follow on or upgrade to an existing software but an entirely new solution to automate, track and monitor the garbage collection process effectively.

The system operates as web application, accessible through standard web browsers on computers and mobile devices. It includes front end interface for users (residents, garbage collectors, and municipal council administrators) and a back end database for storing user data, collection reports, and complaint records.

Garbage collection management system interacts with external systems only for standard web functions such as location tracking. It is otherwise self contained and does not depend on external enterprise system

2.2 Product Features

The Garbage collection management system will provide a set of high level features that support residents, drivers, and municipal council administrators. These features are deigned to improve communication, transparency, and efficiency in waste collection operations

1.User registration and & profile management

- Residents, drivers, and admin officers can create and manage their profiles.
- Role based access ensures that users sees only the functionalities relevant to them.

2.Real time garbage truck tracking

- Residents can view the live location of the garbage truck.
- The municipal council can track garbage truck.
- Reduce uncertainty and prevent missed pickups.

3.Garbage collection scheduling

- Display collection schedules for residents.
- Admin can create, update, and manage garbage pickup schedules.

4.Complaint management system

- Residents can report issues such as missed pickups, overflowing bins, or misconduct.
- Admins can view, manage, and take action on submitted complaints.
- Status updates keep residents informed. (Eg: pending, in progress, resolved)

5.Bin management

- Admin can create new bins, update bin status, assign bins to routes, and manage bin locations.
- Drivers can update bin status after collection. (Eg: emptied, overflowing, not accessible)

6.Route & driver assignment

- Admin can assign routes to trucks and allocate drivers.
- Management of daily operations such as task allocation.

7.Awareness & information sharing

- System provides recycling tips, health guidelines, and updates on environment events.
- Helps educate residents and promote sustainable waste management practices.
- System sends notifications regarding notices

8.Special pickups request management

- Residents can request special pickups.
- Admins can review and approve, reject or reschedule them.

9.QR code bin identification

- Each garbage bin has a unique QR code for unique identification
- Residents and drivers can scan QR code to view bin information and collection status

10.Emergency alert

- Admin can publish emergency alerts
- Alerts appears on residents home page and residents get email notifications

11.Public feedback and rating system

- Residents can rate and optional written feedback.
- Admin and residents can see feedbacks and ratings.

12.Image based garbage type detection (Nice to have function)

- Resident can upload an image of their garbage.
- The system provides information about the type of garbage.

2.3 User Classes and Characteristics

The system will be used by three primary user classes: Residents, Drivers/ Collectors, and Municipal administrators. Each class interacts with different system features based on their role, technical capability, and privilege level.

Residents

Residents are the general public and represent the largest user group of system. They primarily use the platform to track collection trucks, view their area's collection schedule, report issues, and scan QR code & identify bin status and type. Their technical expert is basic, so the interface for this user class should be simple and user friendly. Residents operate with the moderate privilege level and are considered the most important user class, as the system is designed to improve their experience, reduce complaints, and create a cleaner environment.

Drivers/ Garbage collectors

Drivers and garbage collectors use the system to perform their daily operational tasks. They log into view routes, scan QR code and update bin status, and confirm completion of pickups. Their technical knowledge is low to moderate, and the system should provide

clear, minimal steps to update daily activities. Driver use the system should provide clear, minimal steps to update daily activities. Drivers use the system frequently, mostly during working hours and require reliable, easy to navigate interfaces. Their privilege level is moderate, as they work with route data and bin updates but do not manage system wide operations. They are essential for keeping collection information accurate and up to date.

Municipal council administrators

Administrators are responsible for managing and controlling the entire system. They handle account creation for residents and drivers, manage collection schedules, assign routes, update bin details, monitor real time truck movements, process complaints and send email notifications, handle ratings & feedbacks and handle special requests by users. Admin users have high technical familiarity and frequently interact with the system throughout the day. They hold the highest privilege level and have full access to all operations and data. Administrators are a critical user class because the accuracy, reliability, and coordination of the whole system depend on their action and oversight.

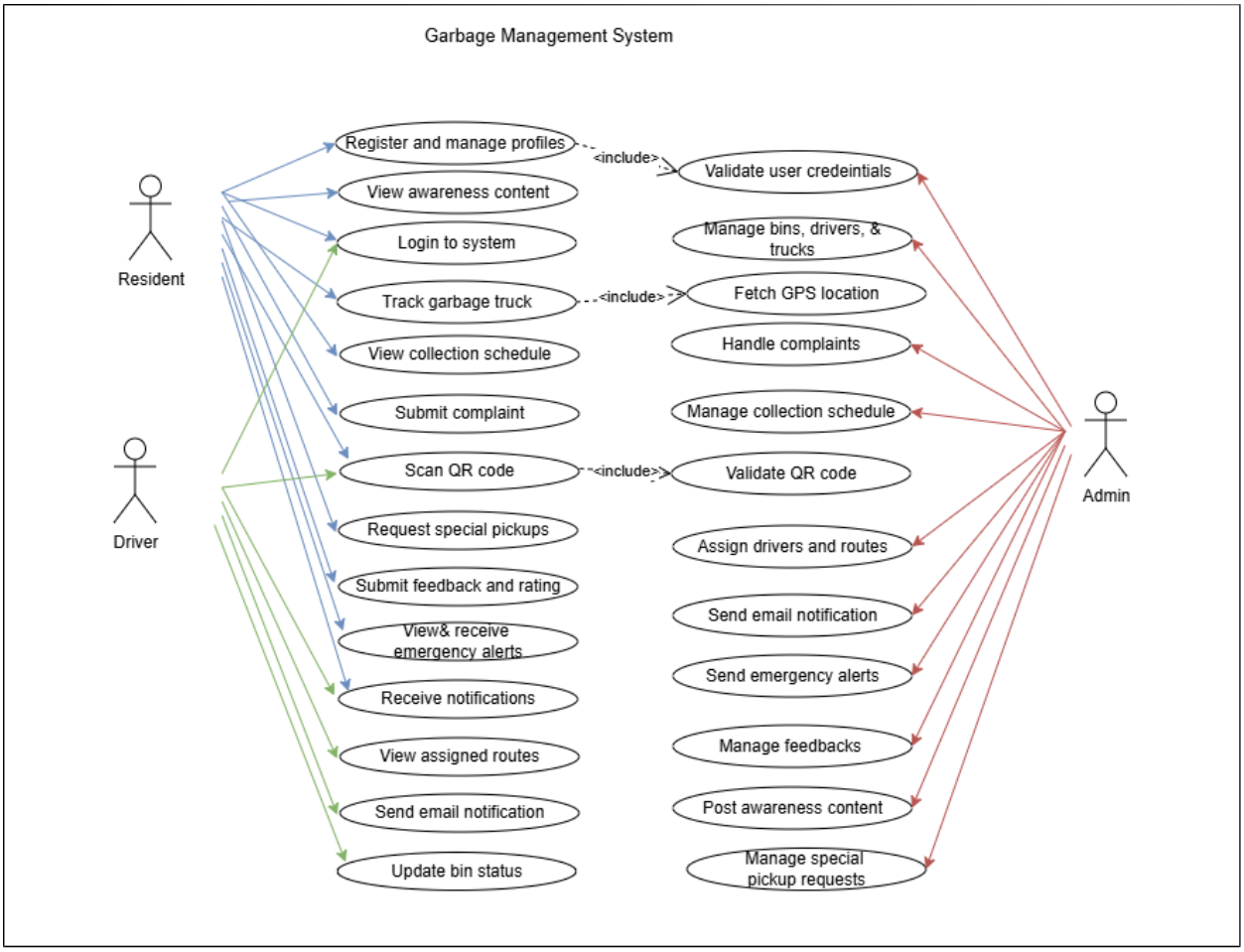


Figure 1: Use case diagram

2.4 Operating Environment

Hardware platform

The system is web based and cross platform, allowing it to run on desktop computers, laptops, tablets, and mobile devices used by residents, drivers and administrators. The server hosting the system should meet the minimum hardware requirements for running the backend, while client devices should support standard web browsing. Any hardware capable of running a modern browser with internet connectivity will be compatible.

Operating system

The system is designed to function on a wide range of operating systems, including windows, macOS, Linux, Android, and IOS. Since different user classes (Residents, Drivers, Administrators) access the system using different devices, the required OS version may vary, but all modern and commonly used versions are supported. The server environment may run on Linux or Windows server, depending on deployment choice.

Software components

The system relies on several software modules to ensure proper functionality. A relational database such as MySQL will be used for storing and managing user data, schedules complaints, and tracking information. Backend development requires React, Node.js, Additional components include Google Maps API for real time truck tracking.

Peaceful Coexistence

All security tools such as antivirus software, firewalls, and network filters must operate without interfering with the system's functionality. The hosting environment and user devices should ensure compatibility with the application, allowing secure connections, map integration, QR code scanning, and database operations without disruption. Proper configuration is required so that these tools coexist smoothly with the garbage collection management system.

2. 5 Design and Implementation Constraints

The system shall be developed using React.js for the frontend, Node.js for the backend, and MySQL as the database, limiting the use of alternative technologies. All communication must occur over secure HTTPS, and external services like Google Maps API. The system requires a stable internet connection for real time features such as truck tracking and QR code scanning. It must also run smoothly on common user devices including smartphones and low spec computers, which limits the complexity of UI elements and processing tasks.

2.6 User Documentation

The following user documents will be provided with the garbage collection management system to support all user groups.

User manual

A complete digital (images and text) user manual explaining system features, login procedures, navigation, and step by step instructions for performing key tasks such as submitting complaints, tracking trucks, and updating bin status.

Quick start guide

A short introductory guide for new users, including screenshots and simple steps to help them begin he system quickly.

2.7 Assumptions and Dependencies

It is assumed that all users including residents, drivers, and administrators will have stable internet access compatible devices to interact with the web based system. The system also relies on third party components such as map APIs for real time truck tracking, meaning any changes or limitations in these service may affect system functionality. It is assumed that the municipal council will provide accurate route, bin, driver information, and will maintain the server environment required for hosting the application. The effectiveness of the system depends on drivers regularly updating bin status and following assigned routes. Additionally, the project depends on the availability of required software tools, such as the selected web framework and the database management system. Any changes in external services, infrastructure, or data availability may impact the final system performance or require adjustments to stated requirements.

3. System Features

3.1 User Registration & Profile Management

3.1.1 Description and Priority

Residents register for the system and create profiles. Users can manage and update their personal information through their profile dashboard.

Priority: High

3.1.2 Stimulus/ Response Sequences

Stimulus: The residents clicks on the “Register” button.

Responses: The registration form is displayed.

Stimulus: The user submits the form with valid registration details.

Responses: System creates the user account, shows a success message, and redirect to the login page.

Stimulus: The user attempts to register with an already used email.

Responses: System displays an error message. “Email already exists”

3.1.3. Functional Requirements

1.The system must allow residents to fill the registration form.

2.The system must validate name, email, address, password and role type

3.The system must prevent registration with duplicate email addresses.

4.After successful registration, users must be redirected to the login page.

Below is an activity diagram to illustrate the flow of user registration functionality in the proposed system.

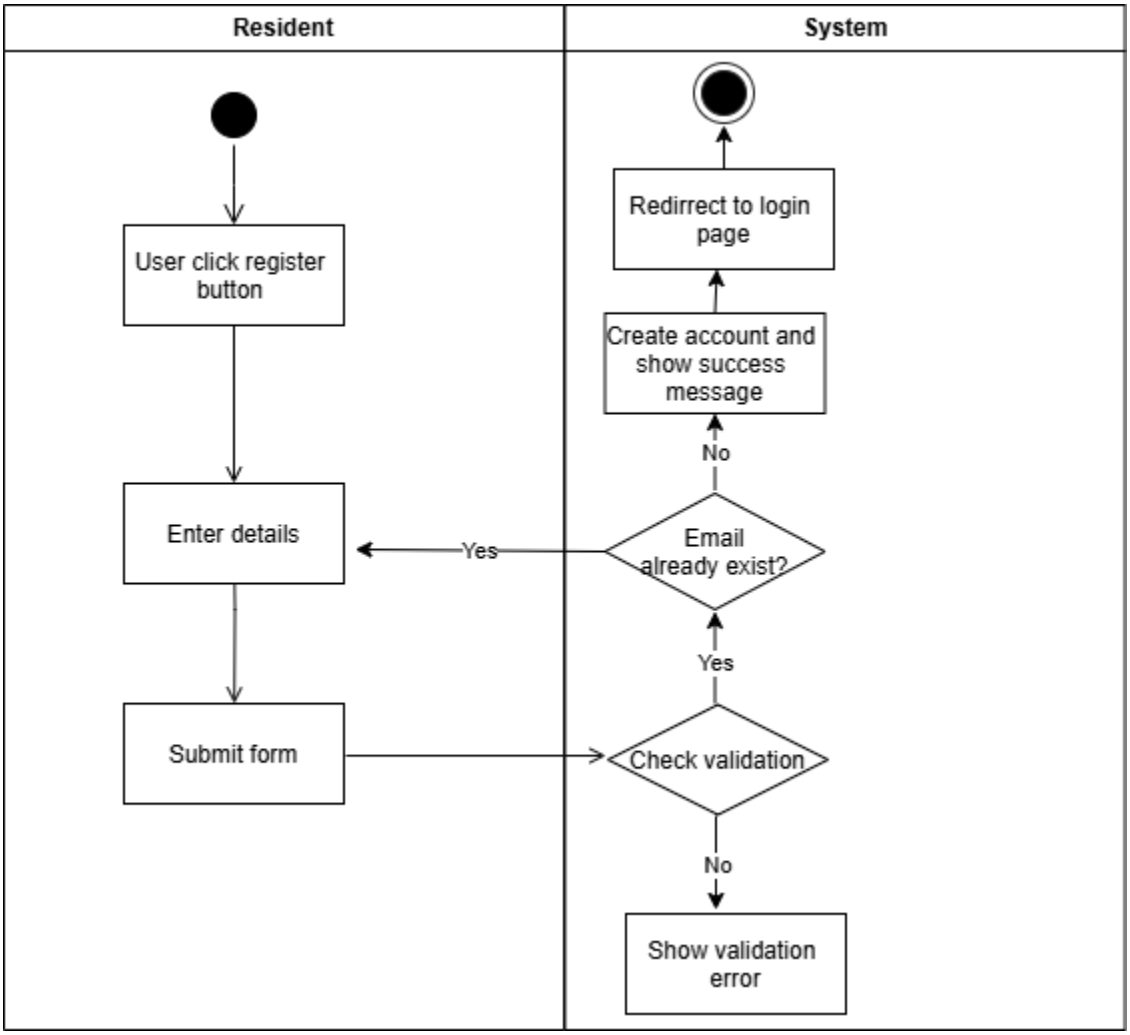


Figure 2: Registration activity diagram

3.2 Login to the System

3.2.1 Description and Priority

All system users (Residents, Drivers, Admin) can log into website and access system features.

Priority: High

3.2.2 Stimulus/ Response Sequences

Stimulus: The user clicks the “Login” button.

Response: Login page displayed.

Stimulus: User **e**nters valid login credentials and clicks submit.

Response: A success message is displayed and the user is redirected to the dashboard.

Stimulus: User inserts invalid login details for three consecutive attempts.

Response: Login is temporarily blocked and notification email is sent.

3.2.3 Functional Requirements

1.Residents, drivers, and admins must be able to fill the login form.

2.The system must validate username and password.

3.The system must redirect each user type to the correct dashboard after logging in

4.After 3 failed login attempts, the system must block login temporarily and send email notification.

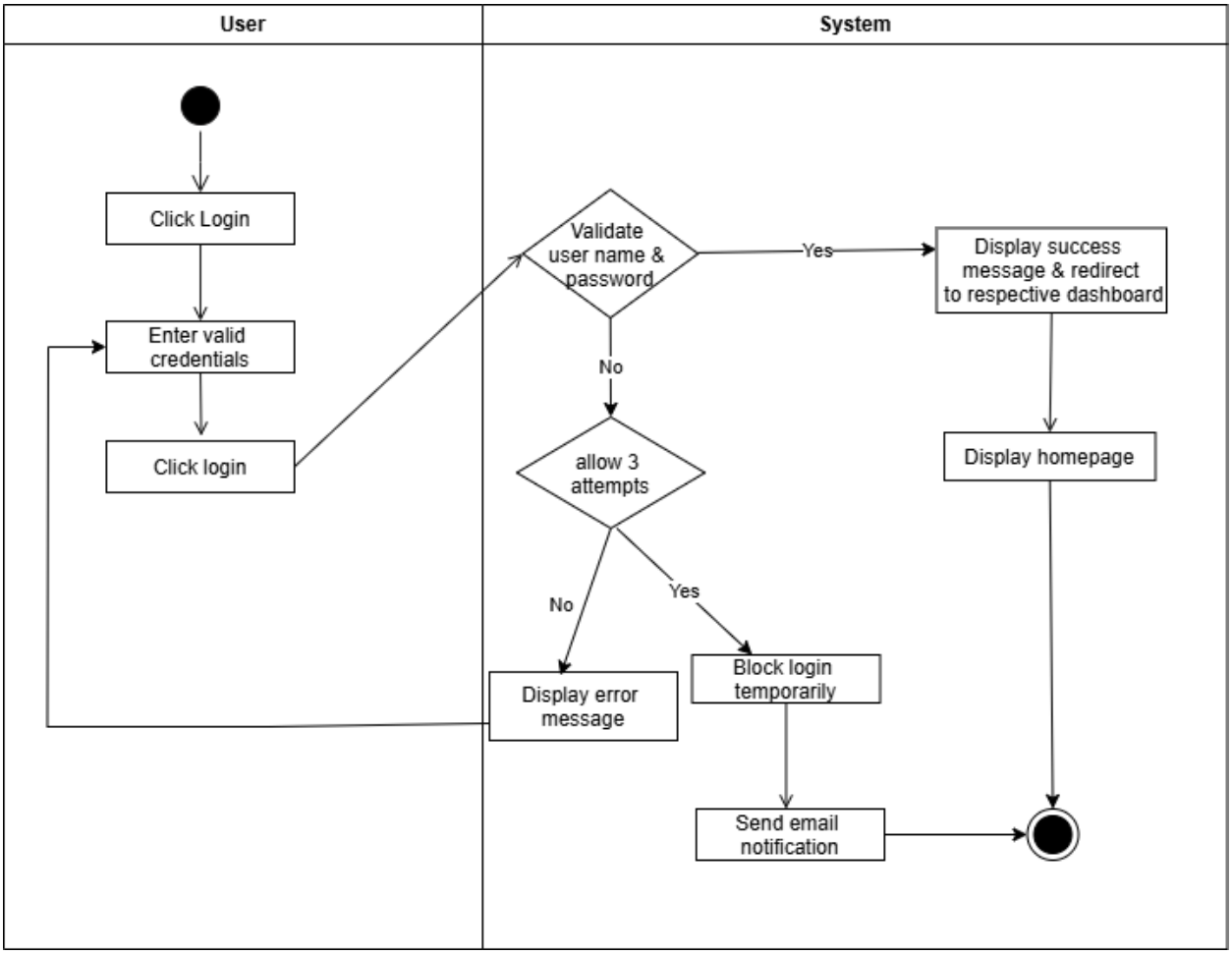


Figure 3: Login activity diagram

3.3 Garbage Truck Tracking

3.3.1 Description and priority

Residents and Admins can track garbage trucks in real time and view estimated arrival times

Priority: High

3.3.2 Stimulus/ Response Sequences

Stimulus: The user selects “Track Garbage Truck”

Response: System loads map and fetches truck GPS data.

Stimulus: Truck location updates in real time.

Responses: System displays live truck movement

Stimulus: GPS device is unreachable or offline.

Responses: System displays “tracking unavailable” message.

3.3.3 Functional Requirements

1. System must collect GPS coordination of garbage trucks.
2. System must update the map in real time.
3. System must show estimated arrival time.
4. System must notify if GPS tracking fails.

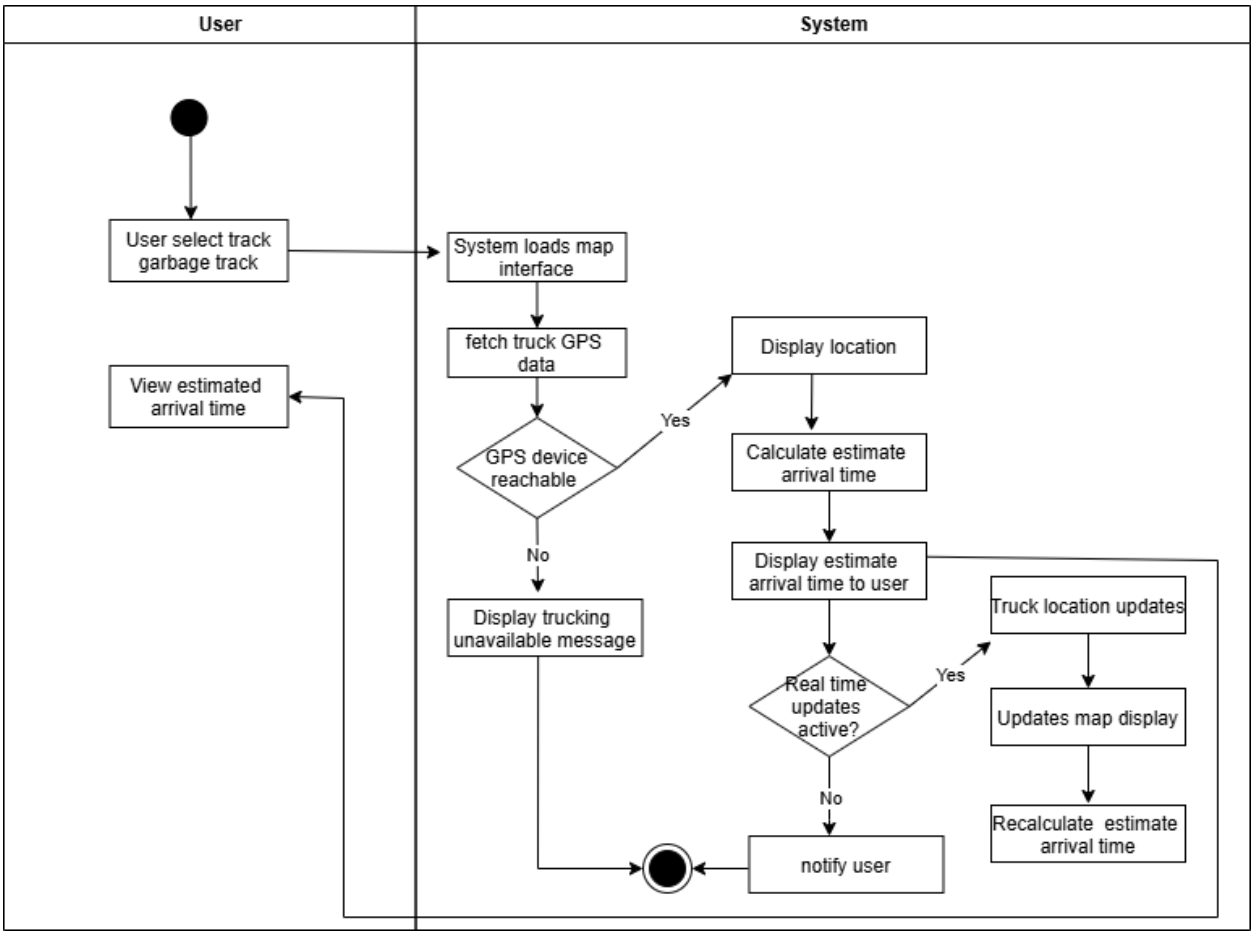


Figure 4: Garbage truck tracking activity diagram

3.4 Garbage Collection Scheduling

3.4.1 Description and Priority

System provides daily and weekly garbage collection schedules for residents and allows admin to modify schedules.
Priority: High

3.4.2 Stimulus/ Response Sequence

Stimulus: Residents clicks “Schedule”

Response: System displays daily schedules.

Stimulus: Admin edits the schedule.

Response: Schedule updates and publishes instantly.

3.4.3 Functional Requirements

1. System must show garbage collection schedules based on resident location.
2. System must allow admin to edit schedules.
3. System must update schedules in real time.
4. System must notify residents about schedule changes
5. .System must store schedule history.

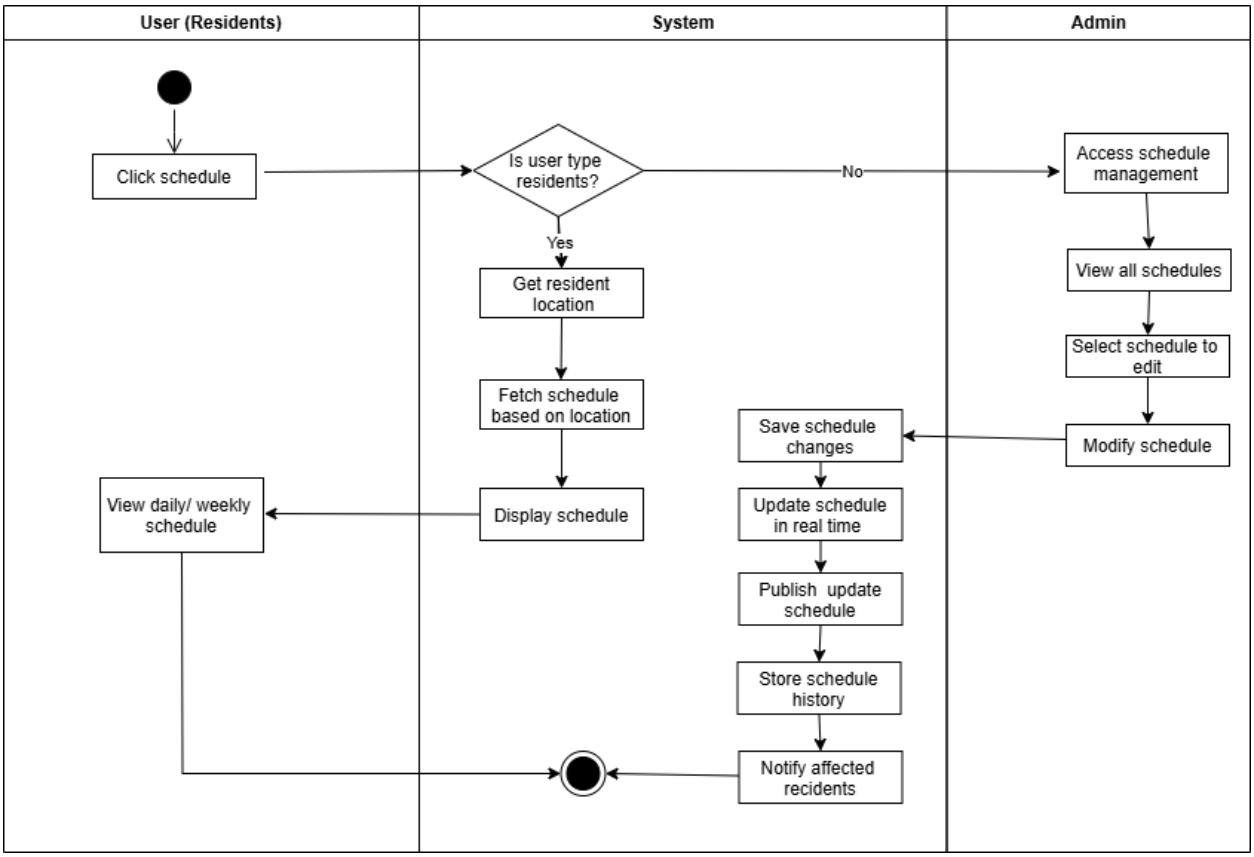


Figure 5: Garbage collection schedule activity diagram

/

3.5 Complaint Management

3.5.1 Description and Priority

Allow residents to submit complaints regarding missed pickups, overflowing bins, or driver behavior, and allows admin to manage resolutions.

Priority: High

3.5.2 Stimulus/ Response Sequences

Stimulus: Resident click “Submit Complaint”.

Response: Complaint form displayed.

Stimulus: Resident submit valid complaint.

Response: System records complaints.

Stimulus: Admin update complaint status.

Response: Resident receives updates on complaints.

3.5.3 Function Requirements

1. System must allow residents to submit complaints.
2. System must validate complaint information.
3. System must allow admin to change complaint status.
4. System must notify residents when complaint progress change.
5. System must store all complaint records.

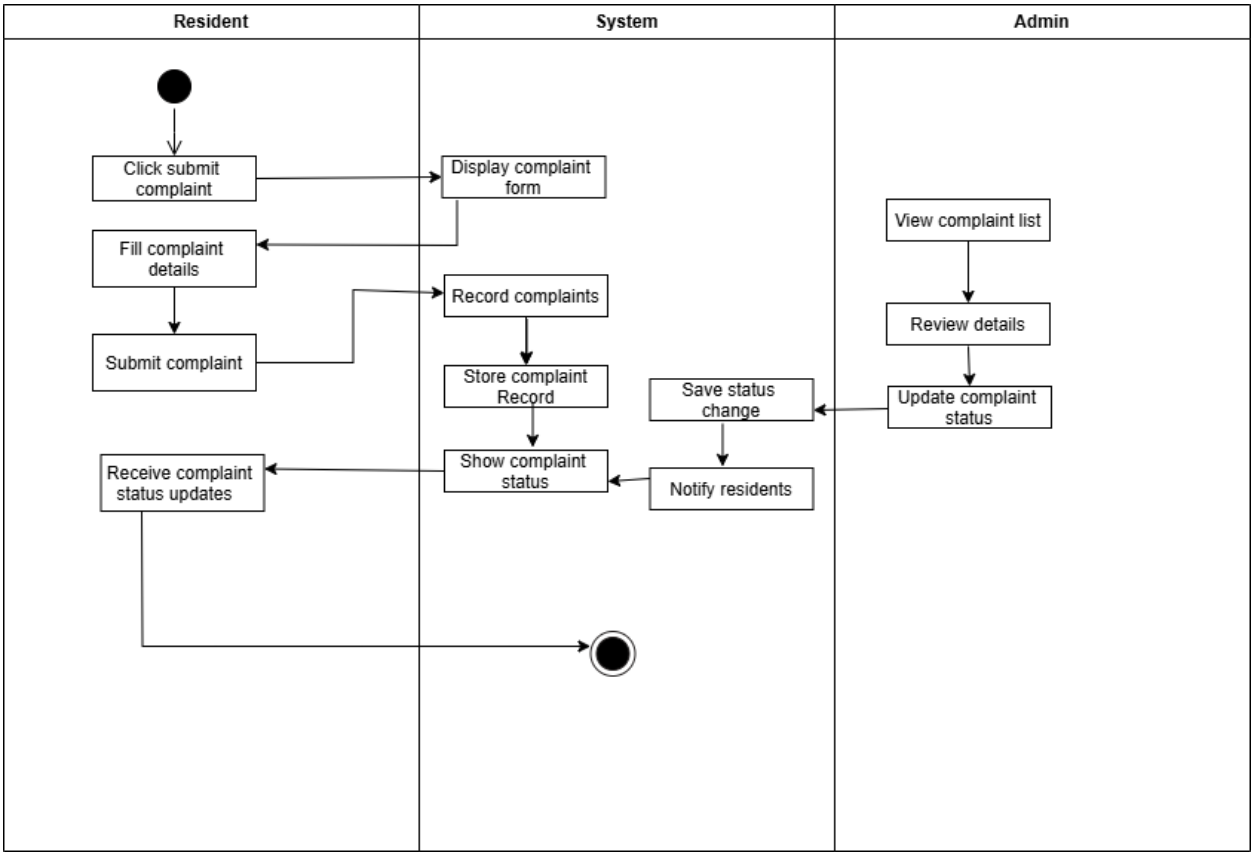


Figure 6: Complaint management activity diagram

3.6 Driver Route & Bin Status Management

3.6.1 Description and Priority

Drivers can view and assigned routes and update the status of garbage bins during collection.

Priority: High

3.6.2 Stimulus/ Response Sequences

Stimulus: Driver logs in and open “Assigned Routes”.

Response: System displays bins and locations.

Stimulus: Driver mark bin as “Collected”.

Response: System records the status.

3.6.3 Function Requirements

1. System must display assigned routes to drivers
2. System must allow drivers to update bin status.
3. System must notify admin for issues like overflow.

3.7 Admin Management (Bins Drivers, Routes, Reports)

3.7.1 Description and Priority

Admin manages bins, drivers, routes trucks, complaints, and system operations.

Priority: High

3.7.2 Stimulus/ Response Sequences

Stimulus: Admin select “Manage Bins”

Response: System displays all bin locations.

Stimulus: Admin assign drivers, routes and bins

Response: System display update new assignments**.**

3.7.3 Function Requirements

1. System must allow admin to add, edit, and delete bins.
2. System must allow admin to assign drivers and trucks.
3. System must allow admins to manage drivers and trucks.

3.8 Public Awareness & Notifications

3.8.1 Description and Priority

System displays recycling guidelines, health information, and sends notifications related to garbage management

Priority: Medium

3.8.2 Stimulus/ Response Sequences

Stimulus: Resident opens “Awareness page”

Responses: System displays articles, tips, and guidelines**.**

Stimulus: Admin posts an event/ announcement.

Responses: System updates awareness page and send notifications.

3.8.3 Functional Requirements

1. System must display segregation and recycling guidelines**.**
2. System must allow admin to upload awareness content.
3. System send notifications for important updates.
4. System must store all published content

3.9 Special Pickup Request Management

3.9.1 Description and Priority

For large items (furniture, construction waste, festival waste), residents can request special garbage pickups. The admin assigns available drivers after reviewing them.

Priority: Medium

3.9.2 Stimulus/ Response Sequences

Stimulus: The resident clicks “Request special pickups”.

Responses: Pickup request form is displayed by the system.

Stimulus: The resident makes a request.

3.9.3 Functional Requirements

1. System must allow residents to submit special pickup requests.
2. System must allow admin to verify the details entered by the residents.
3. Request must be able to be approved, rejected or rescheduled.
4. Residents must be informed of request status.
5. All records of special pickup request status must be stored in the system.

3.10 QR Code Bin Identification

3.10.1 Description and Priority

Each bin has a QR code. Drivers and residents can scan it to view bin information, and check collection status.

Priority: High

3.10.2 Stimulus/ Response Sequences

Stimulus: User scans bin QR code.

Response: System displays bin details**.**

Stimulus: Driver scans bin QR code.

Response: System allows to update bin status.

Stimulus: Driver update bin status.

Response: System update bin history.

3.10.3 Functional Requirements

1. System must generate a QR code for each bin.
2. System must display bin ID, location, and status when scanned.
3. System must allow drivers to update status via QR code.

3.11 Emergency Alerts

3.11.1 Description and Priority

Admin can send emergency alerts related to road closers, or service delays.

Priority: High

3.11.2 Stimulus/ Response Sequences

Stimulus: Admin clicks “Create Alert”.

Response: Alert form displayed.

Stimulus: Residents login to system.

Response: Alert banner displayed in home page.

3.11.3 Functional Requirements

1. System must allow admin to publish emergency alerts.
2. System must display alerts clearly to all users.
3. System must send notifications for alerts.

3.12 Public Feedback and Rating System

3.12.1 Description and Priority

Residents can provide feedback and rating related to the garbage collection service, and admins can view feedbacks.

Priority: Medium.

3.12.2 Stimulus/ Response Sequences

Stimulus: Residents select “Feedback and Rating”

Response: System displays the feedback form with the rating option

Stimulus: Residents **s**ubmits ratings and feedbacks

Response: System records the feedback and shows a confirmation message.

Stimulus: Admin/ residents opens the feedback section.

Response: System displays feedbacks with ratings.

3.12.3 Functional Requirements

1. System must allow residents to submit ratings.
2. System must allow residents to add optional written comments.
3. System must allow admins and users to view feedback

4. External Interface Requirements

4.1 User Interfaces

The following are some wireframes that shows how the interfaces of the application shall appear:

Login - This screen is used to login into the system

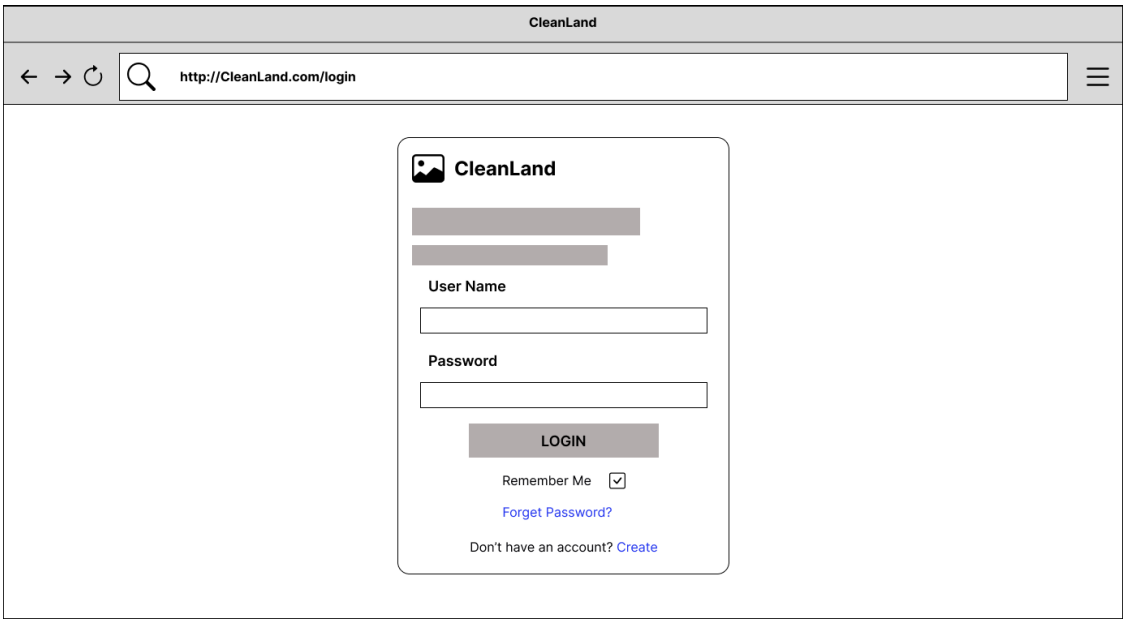


Figure 7: User login interface

Complaint Page

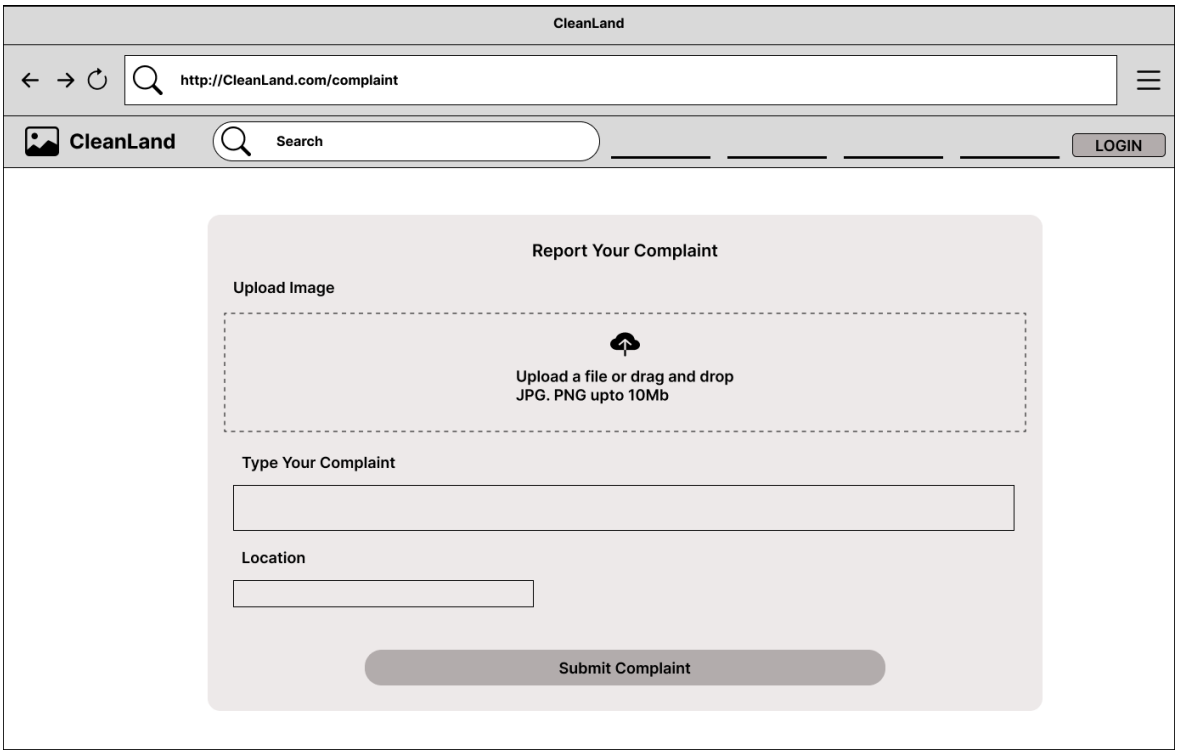


Figure 8: User complaint submission interface

4.2 Hardware Interfaces

This system is a web based application, so smartphone, laptop, or desktop computer are the hardware devices that are needed to connect to the system additionally, the system will be able to work with a wide variety of operating system as this is developed to be cross platform. It will be hosted in a server after deployment

4.3 Software Interfaces

Software and script Components	Name and Version	Description
Operating system	The system is compatible with Windows, macOS, Linux, Android, and iOS through a web browser.	The system will be developed to run on all the operating system
Web browser	Google chrome, Mozilla Firefox	Supported browsers used by residents, drivers and admins to access the system.
Backend	Node.js- Node v24.x	Handles server side logic including authentication, routing, schedule management, complaint handling, and image processing request.
Database	MySQL- v8.4	Stores user accounts schedules, complaints, bin data, and configuration details.
Frontend	React.js- v19.2.0	Used to build the interactive user interface, dashboards, forms, tracking views, and all client-side components.
Mapping Service	Google maps	Provides map rendering.

Table 2: Software interface table

4.4 Communications Interfaces

Communication Function	Description
Email	The system sends email notifications to users.
Browser	Users access the system through a web browser. The frontend is built using React.js and works on Chrome, Firefox, Safari, and Edge.
Electronic Forms	The system provides electronic forms for registration, login, complaints, and other

	actions. Forms send data to the backend through secure REST API calls.
Mapping communication	The system connects to Google Maps to load maps and show truck locations via secure HTTPS calls.
Database communication	Node.js communicates with the MySQL database to store and retrieve all system data using secure queries.
GPS tracking data	Garbage truck location data is sent securely to the backend and displayed on the frontend in real time.

Table 3: Communication interface table

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system should operate smoothly under heavy usage and support multiple users at the same time without slowing down. All main functions, including loading pages, viewing schedules, and tracking garbage trucks should respond within 5 seconds. The server setup must be strong enough to handle a high number of requests from residents, drivers and administrators during peak hours.

5.2 Safety Requirements

The system must protect all stored information and ensure that data is not lost or damaged. Regular backups should be maintained so that older records can be restored if needed. Sensitive operational pages or functions should be accessible only through secure, restricted links to keep the system safe from unauthorized access.

5.3 Security Requirements

The system should ensure that user data is kept secure at all times. This includes using protected storage methods and controlling who can access different parts of the system.

Only authorized users should be able to log in and perform specific actions, ensuring that personal and operational information remains safe from misuses.

5.4 Software Quality Attributes

As the target system is a web-based system and supports any operating system. It is maintainable, adaptable, available, flexible, reliable, and usable.

Maintainable

The system is easy to update and fix because its modules (tracking, complaints, schedules, code scanning) are separated and clearly structured.

Adaptable

The system can work on different devices and can be easily adjusted to new routes, new features, or future system changes.

Available

The web system is accessible anytime through the internet, allowing users to access services like tracking or complaints 24/7.

Flexible

Different user roles (Residents, Drivers, Admins) can use the system with their own features, and ne features or roles can be added easily.

Reliable

The system performs consistently, updates accurate data (GPS, schedules, complaints), and work smoothly even during heavy use.

Usable

The interface is simple and user friendly, so residents, drivers and admins can use the system easily without much training.

6. Other Requirements

6.1 Legal & Regulatory Requirements

1. The system must comply with Sri Lanka’s data protection and privacy laws.

2. User data must not be shared with third parties without consent.
3. System logs and complaint records may be used only for municipal administration purposes.
4. The system must provide secure handling of personal identifiable information.